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Abstract

The administration of multi-vendor classified advertisements is no exception to the central role that digital integration plays in today's corporate landscape. Vendors and customers alike must have access to web interfaces in the classified advertising sector. Creating a digital user interface for a **Classified Multi-Vendor Ads Management Website** is therefore given priority. This article underlines the value of a digital platform in reducing the complex procedures involved in multi-vendor ad management.

The provider-to-consumer aspect of digital ad management is the main topic of this article, which emphasizes the requirement for a seamless, effective, and transparent operation. The project aims to provide an online user interface that gives users access to real-time ad tracking, fast updates, and hassle-free ad insertion and administration. The system has a two-tiered structure with a robust back-end database for data management and user-focused front-end technologies for an easy-to-use interface. It's essential to be proficient in various computer languages, including PHP and Laravel for the back end and HTML, CSS, and JavaScript for the front end. Participants will obtain an in-depth understanding of the coding languages and technologies supporting user interface design and back-end development as they work on this project. This article digs into each underlying technology, illuminating the complexities of creating and implementing a Classified Multi-Vendor Ads Management Website.

Keywords: Classified Ads; Multi-Vendor Platform; Ad Management; User Registration; Payment Processing; Online Payments; User Authentication; User Dashboard; Ad Approval; Search and Filters; Category Management; Responsive Design; PHP Development; Laravel Framework; E-commerce; Digital Advertising; Revenue Generation; SEO Optimization; Mobile Compatibility; Vendor Performance Metrics; Content Management; Classified Listings; Ad Campaigns.

摘要

在当今的企业环境中，数字整合在多供应商分类广告管理中发挥着核心作用。无论是供应商还是客户，都必须能够访问分类广告领域的网络界面。因此，创建一个分类多供应商广告管理网站的数字用户界面成为优先事项。本文强调了数字平台在简化多供应商广告管理中复杂流程的重要性。

本文主要讨论数字广告管理中的供应商与消费者方面，强调了流畅、有效和透明操作的需求。该项目旨在提供一个在线用户界面，使用户能够实时跟踪广告、快速更新以及轻松插入和管理广告。该系统具有双层结构，包括用于数据管理的强大后端数据库和用于易用界面的用户为中心的前端技术。精通各种计算机语言是必要的，包括用于后端的 PHP 和 Laravel，以及用于前端的 HTML、CSS 和 JavaScript。参与者在参与此项目时，将深入了解支持用户界面设计和后端开发的编程语言和技术。本文深入探讨了每项底层技术，阐明了创建和实施分类多供应商广告管理网站的复杂性。

关键词：分类广告；多供应商平台；广告管理；用户注册；支付处理；在线支付；用户认证；用户仪表板；广告审批；搜索和过滤；类别管理；响应式设计；PHP 开发；Laravel 框架；电子商务；数字广告；收入生成；SEO 优化；移动兼容性；供应商绩效指标；内容管理；分类列表；广告活动。

Contents

学士学位论文原创性声明 I
Abstract II
摘要 III
Chapter 1 Introduction 1
1.1 Background Research 1
1.2 Existing Problem 2
1.3 Related Works 3
1.3.1 Existing System 3
1.3.2 Comparative Studies 4
1.4 Purpose of the System 5
1.5 Scope of the Thesis 6
1.6 Key Findings 8
1.7 Structure of the Paper 8
Chapter 2 Literature Review and Conceptual Framework 10
2.1 Digital Classified Advertising 10
2.2 E-Commerce Trends and Marketplaces 11
2.3 Multi-Vendor Platforms 12
2.4 Literature Review on Ads Management System 13
2.5 Advantages and Disadvantages of CMVAMW 14
2.6 Web Development and Hosting 18
2.7 Web-Based Applications 19
Chapter 3 System Analysis and Methodology 21
3.1 System Analysis Objectives 21
3.2 Feasibility Analysis 21
3.2.1 Economic Feasibility 22
3.2.2 Technical Feasibility 22
3.2.3 Social Feasibility 23

3.3 Business Process Modeling (BPM) 23
3.4 Methodology for System Development 24
3.5 System Requirements and Analysis 25
3.5.1 Hardware Requirements 26
3.5.2 Software Requirements 26
3.5.3 Functional Requirements 26
3.5.4 Non-Functional Requirements 27
3.6 Use Case Diagram Analysis 27
3.7 E-R Diagram Analysis 29
3.8 System Architecture Design 31
3.9 Data Flow Diagram Analysis 32
Chapter 4 System and Database Design 35
4.1 User Interface Design Specifications 35
4.1.1 Interaction Design and UI/UX Principles 36
4.1.2 Implementation Requirements 36
4.2 Detailed System Design 37
4.2.1 User System Design 37
4.2.2 Admin System Design 47
4.3 Implementation of the Database 53
4.4 Database Schema and Design 54
Chapter 5 Technology Stack and Implementation 64
5.1 Selected Technology Stack 64
5.1.1 Frontend Technologies 64
5.1.2 Backend Technologies 64
5.1.3 Database Technologies 64
5.2 Frontend Development and Implementation 65
5.2.1 Frontend Code 65
5.3 Backend Development and Implementation 76
5.3.1 Backend Code 76

Chapter 6 System Testing and Report 80
6.1 Testing Strategy and Plan 80
6.2 Functional Testing 80
6.3 Performance Evaluation 80
6.4 Usability Assessment 80
6.5 Security Testing 81
6.6 User Acceptance Testing (UAT) 81
6.7 Testing Results and Comprehensive Report 81
Conclusion 82
References 84
Acknowledgment 87

Chapter 1 Introduction

This chapter introduces the **Classified Multi-Vendor Ads Management Website (CMVAMW)**, a novel digital platform that revolutionizes how classified advertisements from multiple vendors are managed and showcased. In an era marked by increasing digitalization and the proliferation of online marketplaces, there is a growing need for a robust and efficient system to manage classified ads efficiently. The CMVAMW system offers a comprehensive solution to this problem by providing a centralized platform for vendors to manage their ads and for users to search, browse, and interact with them. The chapter opens with background information outlining the existing difficulties and constraints in handling classified ads. The following section lists the field's concerns and examines earlier works that have tried to address related problems. The background provided by this explanation prepares the reader to comprehend the relevance of CMVAMW. The system's goal is explained, highlighting how it may simplify classified ad administration for suppliers and users. The boundaries and emphasis of the research are outlined in the scope of the papers, which also identifies the particular CMVAMW system components that will be discussed in the following chapters. This chapter also summarizes the research's findings and looks at the changes and enhancements CMVAMW makes to the landscape of classified advertisements management. The framework of the paper's structure is provided at the conclusion. It provides a road map for the remaining chapters and their contributions to comprehending and using the Classified Multi-Vendor Ads Management Website.

1.1 Background Research

Classified ads have significantly changed in the digital age, becoming essential to internet business. By bridging the supply and demand gap, they enable interactions between buyers and sellers on multiple platforms. This growth is especially noticeable in multi-vendor e-commerce platforms, where classified advertisements have increased their function in developing various markets for goods and services.[1]

Our thesis, Classified Multi-Vendor Ads Management Website, aims to investigate and solve the opportunities and problems associated with effectively managing and arranging classified

advertisements from numerous suppliers within this dynamic digital environment. This trip demonstrates technology's significant influence on how classified advertising is distributed and accessed today. The rise and widespread use of multi-vendor e-commerce platforms, epitomized by industry behemoths like Amazon, eBay, and Alibaba, have fundamentally changed the online retail environment.[2] These platforms have entirely reimagined how customers find and buy things, highlighting the need for an efficient classified advertisements management system inside such complex ecosystems.

The project must also be in line with user expectations and market conditions.[3] To support the construction of our suggested Classified Multi-Vendor Ads Management Website, I will evaluate market demand through in-depth research and examine user preferences and habits. These revelations will lay the foundation for the following chapters, concentrating on the design, development, and assessment of this cutting-edge platform.

1.2 Existing Problem

Online stores and classified advertisements have permeated every aspect of everyday life in the current digital era. They act as essential marketplaces, bringing together consumers and sellers of goods and services quickly and easily. Nevertheless, there is a severe issue with handling classified advertisements, especially when there are several vendors.

The current issue may be summed up as follows:

- **Fragmented Platforms:** Many classified advertisements and online markets are housed on several platforms, each with its user interface, administration tools, and payment methods. Both vendors and consumers may have a fragmented user experience due to this fragmentation.[4]

- **Vendor Overload:** Multi-vendor marketplaces sometimes have trouble managing a lot of adverts from different suppliers. Due to this overburden, it may become challenging to moderate material, guarantee quality, and offer a consistent user experience.

- **Inconsistent Information:** Inconsistencies in ad information, including prices, product

specifications, or contact data, can erode confidence and result in misunderstandings between consumers and sellers.

- **Limited Visibility:** On traditional classified ad platforms, it may be difficult for smaller vendors or individual sellers to compete with bigger, more established dealers.[5]

- **Security and Trust:** Maintaining safe transactions and fostering trust between buyers and sellers are still issues, especially when fraud is possible.

1.3 Related Works

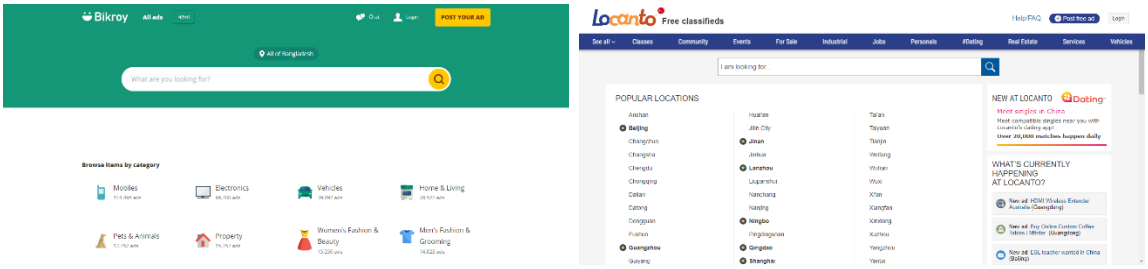
Several solutions have been developed and implemented on a global level in the field of online advertising management systems. The Classified Multi-Vendor Ads Management Website, on the other hand, stands out as a unique strategy developed specifically to answer modern difficulties and improve the entire experience for platform owners and advertisers alike. This system has been carefully designed to gather and handle vendor data and adverts.[6] Its user-friendly online interface ensures that everyone can easily access various information, including administrators and advertising. This solution stands out from competitors due to its consistent focus on effectiveness and usability, especially since it is a web-based platform that simplifies campaign administration for marketers.[7] As a result, I support the usage of the Classified Multi-Vendor Ads Management Website since it not only addresses present problems but also offers a better and more user-centric experience for managing advertising.

1.3.1 Existing System

Several platforms and solutions have arisen in the modern Internet advertising management scene. Platforms that make it easier to put and manage adverts online include "Bikroy.com[8]" and "locanto.cn[9]," among others. These platforms have significantly streamlined the advertising process by removing the need for face-to-face encounters and enabling consumers to conveniently browse available advertising options, make selections, and launch ad campaigns online.

Even while current methods have significantly streamlined the advertising process, they have flaws that need to be acknowledged. In the following parts, we'll go into more detail about these

drawbacks and examine how the Classified Multi-Vendor Ads Management Website seeks to correct and enhance them to provide a more complete and potent advertising management solution.



a) Bikroy.com b) locanto.cn

Figure 1.1 Some Existing Websites

1.3.2 Comparative Studies

The Classified Multi-Vendor Ads Management Website is a comprehensive solution for managing multi-vendor advertising campaigns. This system delivers unique features and functions that set it apart as a better option compared to current platforms offering comparable services. This website's thorough request-handling capabilities are a distinguishing feature. Users may obtain an extensive range of pertinent information about their advertising requests, a component that is noticeably missing from other competing platforms. Additionally, the Classified Multi-Vendor Ads Management Website shines in openness. It gives thorough insights into the status and factors contributing to such choices, unlike other services that may only provide limited information on rejected advertisements.

The system's rigorous documentation and storage of all operations is another critical distinction, making it an essential tool for organizations involved in multi-vendor advertising management. This platform stands out from other advertising management platforms due to its comprehensive record-keeping, which encourages responsibility and efficiency. I will examine these unique characteristics in detail in the following sections and further explain how the Classified Multi-Vendor Ads Administration Website outperforms its rivals in providing a complete and efficient solution for ad campaign administration.

1.4 Purpose of the System

The Classified Multi-Vendor Ads Management Website is conceived to address the multifaceted challenges prevalent within the digital marketplace and online classified advertising domain. A series of pivotal objectives underpin its overarching purpose, each meticulously designed to rectify the existing deficiencies, thereby engendering a transformative effect within the online commerce landscape:

- **Optimized Ad Management:** At its core, the system provides a centralized and intuitive platform whereby multiple vendors can effortlessly create, administer, and refine their classified advertisements. This facet of the system seeks to facilitate the operational experience for sellers, simplifying and expediting the advertisement lifecycle.[10]

- **Elevated User Experience:** Paramount to its raison d'être is the elevation of user experience for sellers or buyers. By imbuing the system with a user-centric design philosophy, it endeavors to endow sellers with an interface characterized by simplicity and efficiency. Buyers, in turn, are poised to benefit from a well-organized, consistent, and engaging browsing milieu, thereby augmenting overall user satisfaction.[11]

- **Vigilant Content Moderation and Trustworthiness:** A foundational element of the system's purpose is ensuring a secure and trustworthy environment. To this end, it incorporates robust content moderation features meticulously designed to enforce content accuracy, reliability, and adherence to community guidelines. This commitment fosters trust amongst users while concurrently mitigating the prevalence of fraudulent or deceptive advertisements.

- **Empowerment of Smaller Vendors:** The system aims to democratize the marketplace, ensuring equitable visibility for smaller vendors and individual sellers. Leveraging intelligent recommendation algorithms and search optimization strategies, it levels the competitive playing field, enabling smaller vendors to contend with larger counterparts on the platform effectively.

- **Secure Transaction Facilitation:** One of the core imperatives of the system is facilitating

secure transactions between buyers and sellers. It is poised to offer accredited and trusted payment processing methods, coupled with streamlined dispute resolution mechanisms, thus diminishing the likelihood of fraudulent transactions and imbuing users with heightened confidence in the platform.

- **Data-Driven Decision Support:** An overarching purpose involves the generation of actionable insights through sophisticated data analytics. The system will aggregate and analyze user behavior, ad performance metrics, and market trends. These insights shall serve a dual purpose: to inform platform enhancements and to furnish vendors with empirically grounded strategies for optimizing their listings.

- **Scalability and Customizability:** As a final overarching objective, the system is meticulously engineered to be inherently scalable and eminently customizable. This flexibility ensures its enduring relevance and adaptability to the evolving dynamics of the online classified advertising ecosystem.

1.5 Scope of the Thesis

Within the ambit of this scholarly endeavor, this collection of papers systematically explores the multifaceted dimensions and intricacies of the "Classified Multi-Vendor Ads Management Website." The scope of these papers is distinctly delineated to provide a comprehensive understanding of the research undertaken and its significance within the domain of online commerce and classified advertising.

The scope of the papers encompasses the following key facets:

- **Conceptual Framework and Problem Definition:** The initial report establishes the foundational conceptual framework, elucidating the pressing challenges and intricacies of managing classified advertisements within a multi-vendor digital marketplace. This paper defines the scope and sets the stage for the research endeavors.

- **Literature Review:** A rigorous examination of extant literature forms the crux of the subsequent paper.[12] This literature review systematically synthesizes existing knowledge,

illuminating the landscape of online classified ads management, multi-vendor platforms, and the underlying challenges. It identifies critical gaps and lays the groundwork for innovative solutions.

- **System Design and Architecture:** The third paper delves into the intricacies of system design and architecture.[13] It meticulously delineates the architectural components, technological underpinnings, and design considerations, culminating in realizing the Classified Multi-Vendor Ads Management Website. This paper articulates the system's blueprint.

- **User Experience and Interface Design:** Central to the research is examining user experience and interface design. This paper scrutinizes the user-centric elements of the platform, elucidating how it enhances usability and fosters positive interactions for both sellers and buyers.

- **Content Moderation and Trust Mechanisms:** The fifth paper is devoted to content moderation and trust mechanisms. It expounds on the strategies to ensure classified advertisements' integrity, accuracy, and compliance. Additionally, it articulates mechanisms instituted to bolster trust and security within the platform.

- **Marketplace Dynamics and User Behavior:** The penultimate paper delves into the online marketplace dynamics and user behavior dynamics. It synthesizes empirical data and insights from the platform's operation, shedding light on emerging market trends and user preferences.

- **Conclusion and Future Directions:** The final paper encapsulates this research endeavor's findings, implications, and overarching contributions. It delineates the practical significance of the Classified Multi-Vendor Ads Management Website and proffers avenues for future research and innovation in the evolving landscape of online classified advertising.

These articles delve deeply into the Classified Multi-Vendor Ads Management Website, examining it from theoretical and practical perspectives. They constitute a comprehensive

exploration, commencing with conceptualizing the issue domain and culminating in the potential overhaul of the online classified ads ecosystem.

1.6 Key Findings

The Classified Multi-Vendor Ads Management Website's essential features and influence on users were examined in a thorough research study to evaluate user experience and system efficacy. The survey results show that users strongly choose the Classified Multi-Vendor Ads Management Website as their top option for managing multi-vendor ad campaigns.[14] The website's user interface efficiently speeds up the ad creation and administration and receives excellent user marks. The report points out issues with current advertising management systems, notably their lack of flexibility in managing multi-vendor ad campaigns.[15] Users complain that working advertising across several platforms is challenging, and they believe the solutions available are limited regarding vendor choice and ad customization.

The Classified Multi-Vendor Ads Management Website successfully tackles these crucial difficulties by providing a uniform and user-friendly platform for managing advertising across a wide range of vendors. The study highlights the website's capacity to offer a more practical and comprehensive ad management experience, enabling users to publish and track their advertising easily. The study emphasizes the relevance of the Classified Multi-Vendor Ads Management Website in addressing changing customer expectations for a comprehensive and adaptable solution to handle their advertising management requirements.

1.7 Structure of the Paper

The paper is distributed into six chapters. Each chapter has various parts explaining in detail.

“Chapter 1: Introduction”, I provide an introduction to the "Classified Multi-Vendor Ads Management Website" thesis. I lay the groundwork by exploring the background and origins of the project, identifying key challenges, establishing objectives, defining scope, and offering a glimpse of initial findings.

“Chapter 2: Literature Review and Conceptual Framework” undertakes a rigorous literature review, focusing on digital classified advertising. I analyze e-commerce trends, multi-vendor platforms, and prior ad management systems, presenting their advantages and disadvantages. Additionally, I discuss web development, hosting, and web-based applications, framing the thesis within the current landscape.

“Chapter 3: System Analysis and Methodology”, I delineate the thesis's system analysis objectives and conduct a comprehensive feasibility analysis, encompassing economic, technical, and social factors. Business process modeling and development methodology guide our approach. We define system requirements, including hardware, software, functional, and non-functional aspects, complemented by visual representations through use cases and E-R diagrams.

“Chapter 4: System and Database Design” delves into system and database design, emphasizing user interface specifications, system architecture, and database schema and structure.

“Chapter 5: Technology Stack and Implementation” outlines the selected technology stack and provides insights into backend and frontend development, offering a practical view of the implementation process.

“Chapter 6: System Testing and Report” focuses on rigorous testing, from strategy and planning to detailed testing results and a comprehensive report, ensuring the system's functionality and robustness.

Chapter 2 Literature Review and Conceptual Framework

2.1 Digital Classified Advertising

Digital classified advertising has experienced a significant transition in the Internet age, altering how companies interact with their target market. Classified advertising has found a new home online after beginning in conventional print media. Geographical restrictions are no longer an issue because of the internet's universal accessibility, enabling advertisers to interact with a larger and more varied audience. As a result of this evolution, dynamic and interactive ad formats have emerged that encourage increased user involvement and higher conversion rates.

Some distinctive traits have evolved in the world of digital classified advertising. These advertisements are placed on websites and online stores, making them instantly accessible and enabling direct communication with potential consumers. Users may still navigate through certain parts or categories since they have retained the categorizing functionality of their print predecessors. Additionally, rich media components like photographs, videos, and interactive material are frequently used in digital classified advertising to improve aesthetic appeal and efficiently communicate information. While direct communication channels, such as contact information and inquiry forms, promote smooth interactions between vendors and possible purchasers; users gain from search and filter choices that help them focus their searches among a sea of postings.

The switch to digital classified advertising has not been without difficulties, especially regarding efficient administration. The sheer number of advertisements from various suppliers can overrun online marketplaces. Robust management solutions are necessary for ensuring ad relevancy, monitoring performance indicators, and preserving a user-friendly experience. Additionally, a complete strategy for ad management is required, given the integration of digital classified advertising with multi-vendor platforms. The suggested Classified Multi-Vendor Ads Management Website is significant in this context since it aims to solve the complexities and demands of this changing marketplace.

2.2 E-Commerce Trends and Marketplaces

E-commerce has seen a revolutionary metamorphosis in recent years, transforming the retail industry. Consumers are increasingly turning to digital platforms due to the accessibility and ease of online buying, which puts a strain on conventional brick-and-mortar retail and changes how companies interact with their clients. The COVID-19 epidemic hastened this transformation further, making e-commerce a necessary component of daily life. Online marketplaces, which have become potent centers of business activity, are at the center of this digital commerce ecosystem. These digital marketplaces, which provide a wide range of goods and services, are created by bringing together various suppliers and customers on these virtual platforms. Industry behemoths like Amazon, eBay, and Alibaba are notable examples, as are specialized websites like Etsy. They have changed the nature of retail and entrepreneurship by democratizing commerce and enabling companies of all sizes to reach a global clientele.

Several significant developments have evolved in the context of this dynamic e-commerce ecosystem. Consumers increasingly select mobile applications and websites for their buying requirements, driving exponential mobile commerce (M-commerce) development. Personalization has become a pillar of e-commerce through customized product suggestions and information powered by data analytics and artificial intelligence. Consumers today anticipate the integration of physical and digital retail channels through omnichannel strategies, and the growing demand for environmentally friendly goods and open supply chains results from worries about sustainability. Additionally, to represent the ever-widening borders of this digital frontier, online marketplaces are extending their reach beyond traditional shopping to include services, subscriptions, digital items, and classified ads.

Effective advertising and ad management are essential in this vast e-commerce environment. Due to the intense competition for consumers' attention, classified ads are necessary for businesses looking to stand out in the online market. However, handling this advertising inside the intricate web of online markets has particular difficulties, such as assuring ad relevancy, monitoring performance indicators, and successfully interacting with people. These difficulties highlight the urgent requirement for an advanced solution like the Classified Multi-Vendor Ads Management

Website. This platform seeks to simplify and improve the management of classified ads, eventually assisting in advancing online shopping and advertising solutions within a dynamic market.

2.3 Multi-Vendor Platforms

Multi-vendor platforms, also known as online marketplaces, have risen to prominence as influential entities in the modern digital realm. These platforms serve as virtual hubs where various vendors gather to exhibit diverse products and services. Their ascent can be attributed to a convergence of factors, including technological advancements, shifting consumer preferences, and the globalization of commerce. Recognizable names like Amazon[16], eBay[17], Alibaba[18], and Airbnb[19] exemplify the diverse niches and markets that these platforms serve. By offering a shared digital marketplace, multi-vendor platforms have fundamentally transformed commerce dynamics, granted vendors of all sizes and backgrounds the opportunity to tap into a vast, often global, customer base. In doing so, they have reshaped the retail landscape and equipped entrepreneurs with the tools needed to thrive in the digital era.

These multi-vendor platforms have unique features that constitute the basis of their operating structure. They stand out above the others for having an extensive range of suppliers representing various sectors and product categories. This diversity encourages healthy competition and guarantees customers access to multiple options. Additionally, these platforms prioritize user-centric interactions, providing simple user interfaces and practical payment methods to boost client happiness. They also enable safe payment processing, protecting buyers and sellers, and frequently include dispute resolution procedures. The prevalence of user-generated evaluations and ratings influences consumer choice and promotes transparency. These systems are also naturally scalable, allowing sellers to grow their customer base and range of goods.

Effective ad management emerges as a critical component to maximize the impact of classified advertising inside multi-vendor platforms. Success in this industry depends on the capacity to keep advertising relevant, track its effectiveness, and successfully engage users, given the fierce competition for consumer attention. As a result, a platform like Classified Multi-Vendor Ads Management Website plays a crucial part in streamlining and improving the administration of classified advertisements within the complex terrain of multi-vendor platforms. As discussed in the

following chapters of this thesis, it aims to substantially contribute to e-commerce and online advertising solutions by achieving this.

2.4 Literature Review on Ads Management System

The inescapable growth of online shopping and the demand for efficient ad management solutions have propelled a remarkable change in the world of digital advertising. This part launches into a thorough literature analysis that delves into the complex world of ad management systems, which serves as the setting for the proposed Classified Multi-Vendor Ads Management Website. Modern digital advertising relies heavily on ad management systems, making the production, distribution, tracking, and optimization of adverts more efficient. Several systems have been developed in response to the increasing complexity of online advertising, which is characterized by a wide variety of advertising platforms, formats, and audience segmentation. As a result, they are essential in assisting companies in navigating this complexity, maximizing the effect of their marketing initiatives, and eventually achieving their goals.

The ability of ad management solutions to simplify ad production and distribution across numerous digital platforms is a fundamental feature. With the help of these tools, marketers may create aesthetically appealing and contextually relevant adverts that appeal to their target demographic. To keep advertisements exciting and compelling, they provide tools for dynamic ad content production, including text, photos, and videos. Systems for managing advertising are excellent at strategic ad targeting and content production. They use advanced targeting algorithms and data analytics to find and target the most pertinent audience segments. This strategy maximizes ad expenditure by reducing wasted impressions and improves marketing efforts' general efficacy and efficiency.

The literature on ad management systems highlights its critical function in digital advertising. These technologies are praised for their capacity to simplify intricate advertising procedures, improve ad targeting, and offer priceless performance information. Existing literature does, however, emphasize the necessity of adaptability and scalability in ad management systems to support the changing digital advertising scene. Ads management systems must be flexible and adaptable as the digital environment changes due to new platforms, technology, and user behaviors.

The suggested Classified Multi-Vendor Ads Management Website fits into this extensive web of ad-management systems. This platform aims to broaden the functionality of ad management systems to the particular context of classified ads inside multi-vendor platforms, providing specialized answers to the problems of ad relevance, performance tracking, and user engagement. The design, implementation, and assessment of this cutting-edge platform will be covered in more detail in the following chapters of this thesis, which will enhance ad management in the dynamic world of online multi-vendor marketplaces.

2.5 Advantages and Disadvantages of CMVAMW

Advantages of Classified Multi-Vendor Ads Management Website:

The Classified Multi-Vendor Ads Management Website offers a multitude of advantages for both buyers and sellers in the online marketplace. These advantages contribute to an enhanced and efficient buying and selling experience, fostering growth and success in the digital commerce ecosystem. Here are some of the key advantages:

For Buyers:

- **Diverse Product Selection:** Buyers benefit from a wide range of products and services offered by multiple vendors on a single platform. This diversity allows them to explore various options and find precisely what they seek.[20]

- **Competitive Pricing:** Multiple vendors foster healthy competition, leading to competitive pricing. Buyers can often find better deals, discounts, and offers on products and services, ultimately saving money.

- **Convenient Shopping Experience:** Buyers can conveniently browse, compare, and purchase products or services from different vendors without visiting multiple websites or physical stores. This streamlines the shopping process and saves time.

- **Trust and Transparency:** Reputable Classified Multi-Vendor Ads Management Websites often incorporate user reviews, ratings, and seller verification mechanisms, instilling trust and transparency in the buying process. Buyers can make informed decisions based on the

experiences of others.

- **Efficient Communication:** Buyers can interact directly with vendors through messaging and inquiry systems, facilitating efficient communication and the ability to seek clarifications or negotiate terms before making a purchase.

- **Advanced Search and Filtering:** Users can employ advanced search and filtering options to quickly locate products or services based on their preferences, narrowing down choices based on price range, location, or specific features.

- **Personalized Recommendations:** The platform may utilize algorithms to provide personalized product recommendations to users based on their browsing and purchase history, enhancing the shopping experience.

For Sellers:

- **Access to a Wider Audience:** Sellers gain access to a more extensive and diverse customer base by listing their products or services on a Classified Multi-Vendor Ads Management Website, expanding their market reach.[21]

- **Cost-Effective Advertising:** Online classified ads are often more cost-effective than traditional advertising methods. Sellers can reach potential customers without the high costs associated with print or television advertising.

- **User-Friendly Ad Management:** The platform typically offers user-friendly tools and interfaces for creating, managing, and optimizing classified advertisements. This simplifies the advertising process, even for those with limited technical expertise.

- **Data-driven insights:** Sellers can access valuable data and analytics on ad performance, audience behavior, and market trends. This data enables them to refine their advertising strategies for better results.

- **Flexibility and Control:** Sellers control their advertisements' content, pricing, and timing. They can modify and update listings to adapt to changing market conditions.

- **Customer Feedback:** Feedback and ratings from buyers can help sellers improve their products or services, build trust, and enhance their online reputation.

- **Efficient Inventory Management:** Sellers can efficiently manage their inventory and track product availability, ensuring accurate listings and reducing the risk of overselling or underselling.

Disadvantages of Classified Multi-Vendor Ads Management Website:

While Classified Multi-Vendor Ads Management Websites offer numerous benefits, they also come with certain disadvantages and challenges that users, both buyers and sellers, may encounter. It's essential to be aware of these potential drawbacks when using or operating such platforms. Here are some of the disadvantages:

For Buyers:

- **Quality and Trust Issues:** As these platforms host a wide range of vendors, buyers may encounter varying product or service quality levels. Trusting unknown vendors can sometimes be risky, mainly if no precise quality control mechanisms exist.

- **Scams and Fraud:** Some listings on these platforms may be fraudulent or deceptive. Buyers may encounter scams or counterfeit products, potentially resulting in financial losses.

- **Overwhelming Choices:** The extensive variety of products and services can overwhelm buyers. It may be challenging to navigate and decide when presented with too many options.

- **Inaccurate Listings:** Some may contain inaccurate or outdated information, such as incorrect pricing or product descriptions. This can lead to misunderstandings and disputes between buyers and sellers.

- **Communication Challenges:** Interacting with multiple vendors can become time-consuming, and communication issues may arise. Buyers may face delays or difficulties in responding to inquiries or resolving issues.

- **Privacy Concerns:** Buyers may need to share personal information, including contact details and payment information, with multiple vendors, raising concerns about data privacy and security.

For Sellers:

- **Increased Competition:** While the platform provides access to a broader customer base, sellers face heightened competition from other vendors offering similar products or services. This can lead to pricing pressures and reduced profit margins.

- **Platform Fees:** Many Classified Multi-Vendor Ads Management Websites charge fees or commissions for listing products or services, which can add up, affecting a seller's overall profitability.

- **Ad Visibility and Ranking:** Advertisements may not always receive prominent placement or visibility, depending on the platform's algorithms or the level of competition. This can impact the number of potential buyers who see the listings.

- **Customer Feedback:** Negative feedback or ratings from buyers can harm a seller's reputation and impact their ability to attract new customers. Managing and responding to feedback is crucial but can be time-consuming.

- **Transaction Risks:** Sellers may face the risk of fraudulent buyers or payment disputes, resulting in financial losses or the need to deal with complicated dispute resolution processes.

- **Inventory Management:** Managing inventory and ensuring accurate listings can be challenging, especially for sellers with an extensive and constantly changing product catalog.

- **Advertising Costs:** While digital advertising can be cost-effective, sellers may need additional advertising to ensure their listings stand out among competitors, incurring extra expenses.

2.6 Web Development and Hosting

Web Development: Web development is the process of building and managing websites and online applications.[22] It is a multidisciplinary area that uses a variety of computer languages, frameworks, and technologies to realize digital concepts. The responsibility of creating dynamic, user-friendly web experiences falls to web developers. Web development's essential elements include:

- **Front-End Development:** Front-end developers focus on a website's visual and interactive aspects. They design the user interface with HTML, CSS, and JavaScript, ensuring it is responsive, aesthetically pleasing, and functional across different devices and browsers.

- **Back-End Development:** Back-end developers build the server-side infrastructure that powers websites and applications. They work with server-side scripting languages (e.g., PHP, Python, Ruby) and databases to handle data storage, user authentication, and the server's logic.

- **Full-Stack Development:** Full-stack developers possess expertise in both front-end and back-end development. They have a holistic understanding of web technologies and can work on all aspects of a web project.

- **Web Frameworks and CMS:** Developers often leverage web frameworks (e.g., Ruby on Rails, Django) and content management systems (e.g., WordPress, Drupal) to streamline development and accelerate project timelines.

- **Security:** Web developers must be vigilant, implementing measures to protect websites and user data from cyberattacks and breaches.

Hosting: Providing server space, infrastructure, and services to enable websites and web applications to be accessed through the internet is called web hosting.[23] It entails placing website assets like databases, files, and other resources on servers and making them reachable to users via a domain name. Essential elements of web hosting include:

- **Types of Hosting:** Various types of hosting are available, ranging from shared hosting

(multiple websites share server resources) to dedicated hosting (an entire server dedicated to one website). The choice of hosting type depends on factors like website traffic and resource requirements.

- **Server Location:** The geographical location of the hosting server can impact website performance and loading times. Proximity to the target audience is often considered for optimal user experiences.

- **Scalability:** Scalable hosting solutions enable websites to accommodate increased traffic and resource demands. This is essential for platforms like the "Classified Multi-Vendor Ads Management Website" that may experience variable loads.

- **Uptime and Reliability:** High uptime and reliability are critical for ensuring that websites are accessible to users around the clock. Web hosting providers often offer Service Level Agreements (SLAs) to guarantee uptime percentages.

- **Security Measures:** Hosting providers implement security measures to safeguard server infrastructure and protect websites from common threats, including DDoS attacks, malware, and data breaches.



Figure 2.1 Web Development and Hosting

2.7 Web-Based Applications

Web-based applications, often web apps, are vital to the digital world and influence how we engage with software and services online.[24] These web-based apps provide various benefits that have completely changed how people and organizations interact with technology. In particular, web applications are well known for their usability and cross-platform compatibility, eliminating the need for complicated installs and allowing users to use them from various devices equipped with web browsers and internet connectivity. This adaptability makes users more comfortable and broadens the range of applications.[25]

The client-server concept lies at the heart of web-based application architecture. In this paradigm, the user's web browser serves as the client and communicates with a distant server, which hosts the application. This approach streamlines the administration of software resources by enabling centralized data storage, seamless upgrades, and effective maintenance. Responsive design principles ensure that web apps offer consistent and user-friendly experiences across various screen sizes and device kinds, considering user preferences.

The Classified Multi-Vendor Ads Management Website is a web-based application that simplifies the management of classified ads on multi-vendor platforms. It addresses security and data integrity issues, protecting against cyber-attacks and preserving user data. The application's practical implementation and factors involved in creating and deploying it will be discussed in later chapters. Web-based apps are known for their real-time interactivity, enabling quick collaboration, communication, and data changes. These capabilities benefit applications like chat platforms, team collaboration tools, and data tracking systems. The web app's scalability and maintenance are also significant advantages, as its architecture simplifies scaling up to handle user growth and changing features. The server serves as the primary location for updates and maintenance.

Chapter 3 System Analysis and Methodology

3.1 System Analysis Objectives

The system analysis phase plays a crucial function with several goals in creating the Classified Multi-Vendor Ads Management Website.[26] In this phase, the system requirements are thoroughly gathered and painstakingly documented, the problem domain is precisely defined, the functional and non-functional needs are articulated, and the relevant system constraints are acknowledged. Engagement with various stakeholders, including users, vendors, and administrators, is crucial throughout this phase. The objective is to glean and internalize a thorough awareness of their requirements and expectations. This necessitates a thorough investigation of the difficulties and inefficiencies pervasive in the field of managing classified ads on multi-vendor platforms. While non-functional requirements are developed to cover crucial aspects like performance, security, and usability, clear and detailed functional requirements are established to define system features and interactions precisely. Furthermore, it is crucial to carefully analyze any limits and limitations impacting how the system is designed and put into practice. In a sense, the system analysis phase acts as the cornerstone, offering a painstakingly laid out roadmap for the system development phases that follow, including design, implementation, and stringent testing procedures.

3.2 Feasibility Analysis

Feasibility analysis is critical in developing the Classified Multi-Vendor Ads Management Website.[27] It serves as a comprehensive evaluation framework for assessing the viability and practicality of the proposed system from multiple perspectives, including economic, technical, and social feasibility. This multifaceted analysis is essential to inform decision-making and ensure that the project aligns with its intended objectives and the broader context of its implementation.

3.2.1 Economic Feasibility

Economic feasibility focuses on the financial viability of the project. It involves a rigorous examination of the costs and benefits associated with developing and operating the Classified Multi-Vendor Ads Management Website. Key considerations include:

- **Cost Estimation:** This entails identifying and quantifying the costs associated with system development, such as software development, hardware infrastructure, personnel, and ongoing operational expenses.

- **Benefit Analysis:** Evaluating the potential benefits is equally crucial. These may include increased revenue from advertising, cost savings through efficient ad management, and possible revenue-sharing models with vendors.

- **Return on Investment (ROI):** ROI calculation helps stakeholders assess whether the expected benefits outweigh the costs incurred. It provides a quantitative measure of the project's economic viability.

- **Risk Assessment:** Identifying potential financial risks, such as cost overruns or revenue shortfalls, is essential for risk mitigation planning.

3.2.2 Technical Feasibility

Technical feasibility assesses the system's ability to be developed and implemented successfully from a technical standpoint. Key factors to consider include:

- **Technology Stack:** Evaluating whether the chosen technology stack and tools align with the project's requirements and objectives.

- **Infrastructure:** Assessing the availability and scalability of the required hardware and network infrastructure to support the system's performance and user load.

- **Development Expertise:** Evaluating the availability of skilled developers and technical expertise required for the project. This includes considering the accessibility of specialized

skills in web development, database management, and security.

- **Integration:** Ensuring the system can seamlessly integrate with existing technologies, databases, and third-party services, if applicable.

3.2.3 Social Feasibility

Social feasibility focuses on the impact of the system on users and the broader community. Key considerations include:

- **User Acceptance:** Assessing the willingness and enthusiasm of users, including vendors, advertisers, and administrators, to embrace and utilize the Classified Multi-Vendor Ads Management Website.

- **User Experience:** Ensuring the system is designed with a user-centric approach, with features and interfaces that align with user preferences and needs.

- **Ethical and Social Implications:** Identifying potential ethical and social implications related to privacy, data security, and the impact of targeted advertising on users.

- **Community Impact:** Evaluating how the system may affect the broader community, including its potential to create economic opportunities for vendors and contribute to the overall digital ecosystem.

3.3 Business Process Modeling (BPM)

Business Process Modeling (BPM) is a significant methodological technique in constructing the Classified Multi-Vendor Ads Management Website. BPM uses graphical notations, sometimes flow diagrams, to clarify the complex system operation. This strategy accomplishes the critical task of giving a clear knowledge of the operational complexities of the system by representing internal business processes in a way that is both visual and understandable. BPM is a unifying language that transcends technical complexity, facilitating efficient communication among stakeholders, including technical specialists, business analysts, and end users. BPM expedites understanding and supports decision-making, both of which are essential for the efficiency and effectiveness of corporate

operations. It provides a brief and visual representation of the complete business activity. To ensure that the system is technically sound and in line with the overarching objective of improving user experiences and optimizing ad management within the multi-vendor ecosystem, BPM will be a key component of the proposed website's modeling and optimization of ad management, user interactions, payment processing, and administrative tasks.



Figure 3.1 BPM for Classified Multi-Vendor Ads Management Website

3.4 Methodology for System Development

The System Development Life Cycle (SDLC), a comprehensive framework with various stages, including Planning, Analysis, Design, and Implementation, was chosen as the development technique for the Classified Multi-Vendor Ads Management Website. The Waterfall Development model is the organized design technique used within this overall SDLC framework. The sequential progression of the waterfall model, in which each step must be finished before going on to the next, is its defining feature. This method offers a path for system development that is well-organized and thoroughly evaluated, precisely aligned with the project's goals. It ensures that requirements, design

considerations, and implementation phases are thoroughly investigated, finally supporting the effective deployment of the website to satisfy end customers' expectations.

The Waterfall methodology and SDLC were specifically chosen to provide a planned and meticulous development approach for the Classified Multi-Vendor Ads Management Website. This method ensures a systematic evolution, with each phase being rigorously implemented and confirmed before moving on to the next. It is especially well-suited for the challenges of developing a multi-vendor advertisements management platform. This process ensures that the development moves forward with clarity and accuracy, finally leading to effectively achieving the website's goals and satisfying its users' demands.

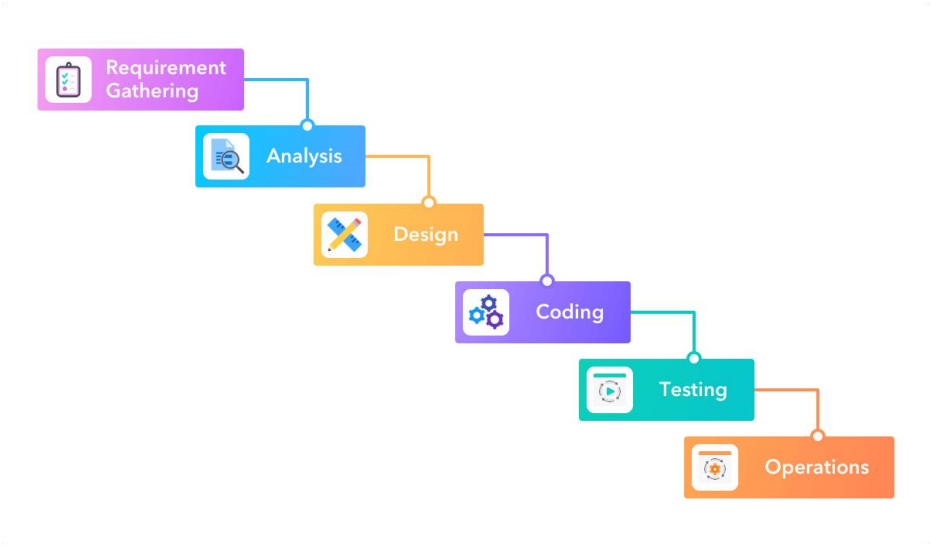


Figure 3.2 Waterfall Model of Classified Multi-Vendor Ads Management Website

3.5 System Requirements and Analysis

The System Requirements phase for the Classified Multi-Vendor Ads Management Website involves defining software and hardware needs, functionality, and non-functional aspects such as performance, security, usability, and scalability. This ensures the system meets stakeholder expectations and delivers optimal performance and user experiences.

3.5.1 Hardware Requirements

- **Processor:** Intel Core i5 or equivalent
- **RAM:** 16 GB (DDR4)
- **Storage:** 1TB SSD
- **Server:** Robust server infrastructure
- **Network:** High-speed internet with load balancing
- **Security:** Advanced firewall and intrusion detection

3.5.2 Software Requirements

- **Operating System:** Windows
- **System Design:** Figma, Adobe Photoshop
- **Programming Languages:** HTML, CSS, JavaScript, PHP
- **Framework:** Laravel, Bootstrap 4 (CSS)
- **Code Editor:** Visual Studio Code
- **Devices:** Laptop
- **Testing:** Laptop, Mobile, Desktop

3.5.3 Functional Requirements

- **User Registration:** Allow user account creation.
- **User Authentication:** Provide secure login and sign-out.
- **Request Approval:** Validate vendor and ad requests.

- **Maintenance:** Ensure ongoing system accuracy.
- **User Dashboard:** Enable profile management.
- **Real-time Data:** Provide ad performance metrics.
- **Scheduling:** Allow ad placement and management.

3.5.4 Non-Functional Requirements

- **Performance:** Ensure fast response times.
- **Efficiency:** Optimize memory and resource usage.
- **Speed:** Quick page and content loading.
- **UI Design:** User-friendly and visually appealing.
- **Security:** Robust data protection measures.
- **Scalability:** Handle increased user traffic.
- **Reliability:** High uptime and stability.
- **Accessibility:** Adherence to accessibility standards.
- **Compatibility:** Support multiple browsers and devices.
- **Data Backup:** Regular data backups for security.

3.6 Use Case Diagram Analysis

A use case diagram, created by the Unified Modeling Language (UML) architecture, is used to depict the functionality and interactions of the system visually. It offers a thorough understanding of the system's operational scope through the graphical representation of its functions and responsibilities. Various user groups, including vendors, marketers, and administrators, are represented by actors in the diagram who play crucial roles and have different

actions and duties. These use cases outline the main scenarios and functionality of the website's interactions with different entities. We may better grasp the Use Case Diagram's function and purpose by gaining useful insights into the roles and activities within the multi-vendor advertisements management ecosystem. The Use Case Diagram may not fully capture the complete system procedure.

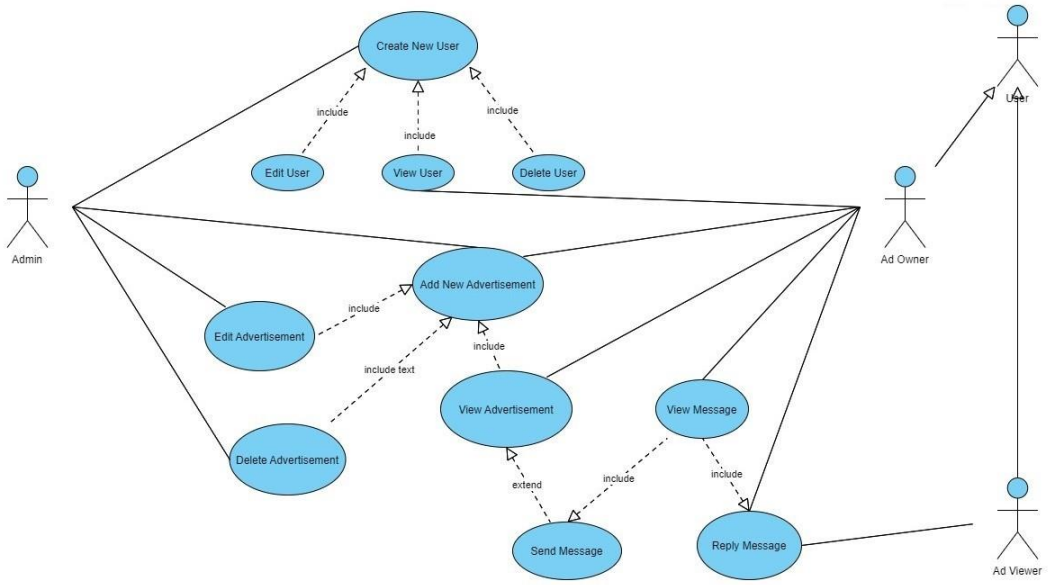


Figure 3.3 Use Case Diagram of Classified Multi-Vendor Ads Management Website

User Activities and Scenarios:

- **Registration:** Users sign up, providing essential details for full access.
- **Login:** Registered users log in for personalized features and ad management.
- **Add Advertisement:** Vendors create and publish ads with details, images, and pricing.
- **View Advertisement:** Users browse, search, and view ads, filtering and accessing details.
- **Messaging:** Users can communicate about ads and services on the platform.
- **Account Management:** Update personal info, passwords, and notification preferences.

- **Customer Support:** Access customer support via live chat, email, or helpline.
- **Feedback & Ratings:** Users rate ads and vendors, fostering trust and transparency.

Admin Activities and Scenarios:

- Admin logins to the system and manage all the functionalities of the Classified Management System
- Admin can add, edit, delete, and view the records of Advertisement, User, Create Registration
- Admin can manage all the details of Location, Categories, Listing
- Admin can also generate reports of Advertisement, Location, User, Categories, Create, Listing
- Admin can search the details of Location, Create, Listing
- Admin can apply different levels of filters on the report of Advertisement and categories. Create
- Admin can track the detailed information of Location, User, Categories, Create

3.7 E-R Diagram Analysis

The Classified Multi-Vendor Ads Management Website's Entity-Relationship (E-R) Diagram visually depicts the key entities and connections inside the system. This blueprint illustrates the relationships between several database tables, including Vendor, Advertisement, User, Category, and more. The links between these crucial elements of the Multi-Vendor Ads Management ecosystem are defined using structured data.

The system's main entities are vendor, advertisement, user, category, and transaction. These elements work together to construct the website's framework, and they are related by relationships that the E-R Diagram painstakingly captures. The Classified Multi-Vendor Ads Management

Website's underlying structure and the critical role each entity plays in enhancing the system's overall functionality can be thoroughly understood by stakeholders by carefully examining this diagram, demonstrating how these entities interact.

Classified Multi-Vendor Ads Management Website's entities and their attributes:

- **User Entity:** User attributes are user_name, #user_id, user_mobile, user_address, user_email, user_pass.
- **Advertising Entity:** Advertisement attributes are advr_place, #advr_id, advr_desc, advr_title, advr_type.
- **Location Entity:** Location attributes are loc_time, loc_place, loc_user_id, loc_desc, #loc_id.
- **Listing Entity:** Listing attributes are #list_id, list_desc, list_type.

Description of Classified Multi-Vendor Ads Management Website's Database:

- The database includes tables for Ads, Vendors, Categories, Users, and Listings, each with unique primary keys.
- Relationships between Vendors, Ads, Listings, Users, and Categories are established using foreign keys for efficient data management.
- The database uses one-to-one and one-to-many relationships for effective organization and data retrieval.
- Data is normalized to minimize redundancy and enhance data integrity.
- Tables are indexed to optimize query performance and ensure a responsive system.

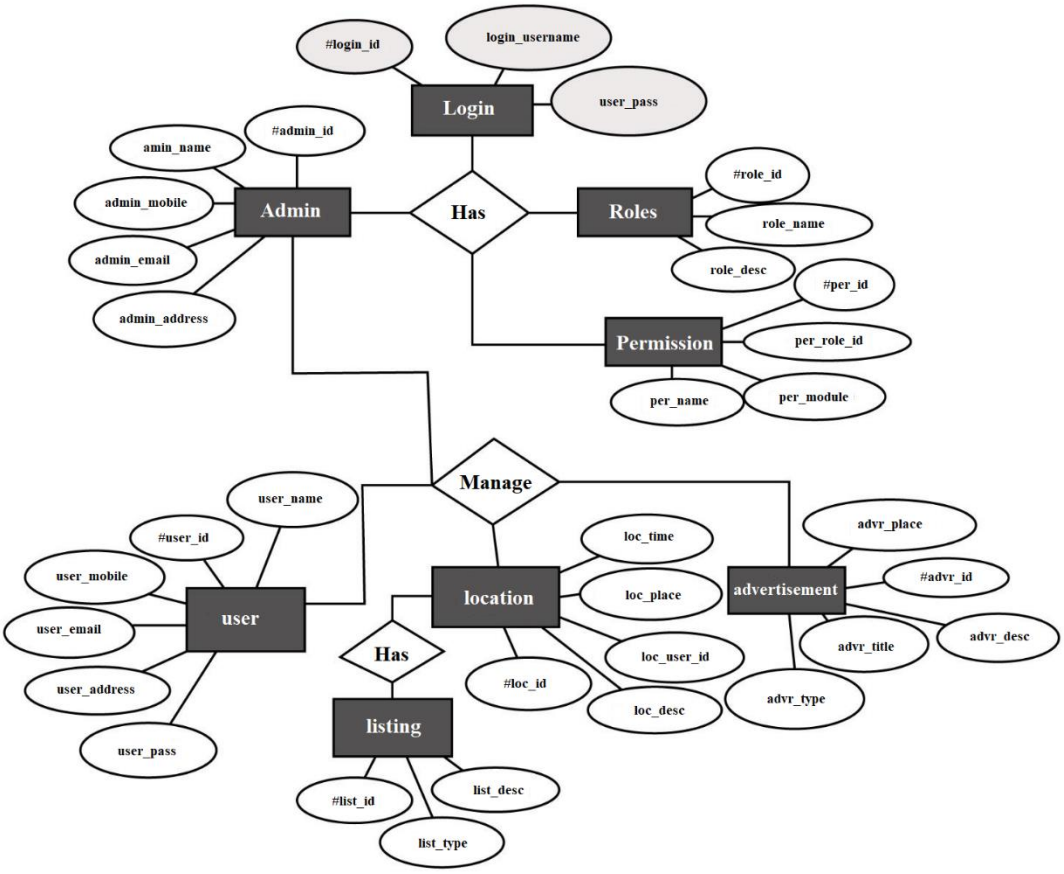


Figure 3.4 E-R Diagram of Classified Multi-Vendor Ads Management Website

3.8 System Architecture Design

The System Architecture Design for the Classified Multi-Vendor Ads Management Website is meticulously crafted to align with existing infrastructure requirements and anticipate evolving user needs. The design encompasses a three-tier architecture:

i. **User Interface:** This tier provides the front-end experience for users, including vendors, advertisers, and administrators. It offers a user-friendly interface for ad management, profile customization, and interaction with the system's features.

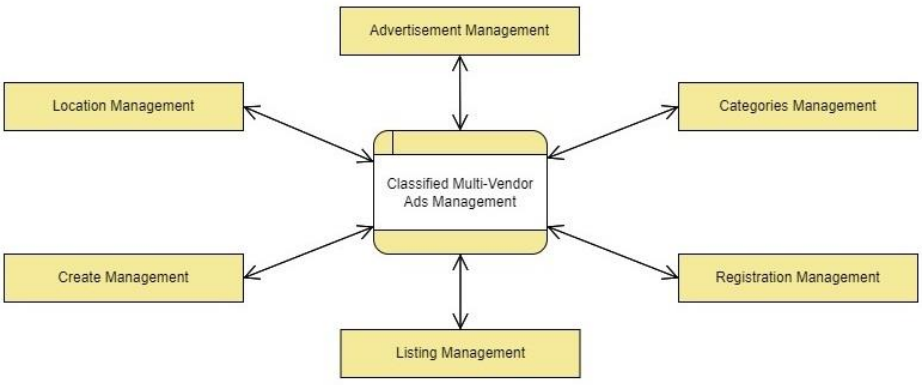
ii. **Process Management:** At the system's heart, this tier handles core functionalities such as ad placement, user authentication, and data processing. It ensures seamless communication between the user interface and the database, facilitating efficient ad management.

iii. **Database Management System (DBMS):** The database tier stores and manages critical data, including user profiles, ads, and transaction records. It is pivotal in ensuring data integrity, scalability, and retrieval efficiency.

This system's components, services, and connections are highlighted by its three-tier design, ensuring top performance and reliable communication between system modules. The Classified Multi-Vendor Advertisements Management Website is prepared to offer a scalable, effective, and user-centric platform for managing classified advertisements inside the multi-vendor ecosystem by adhering to its architecture.

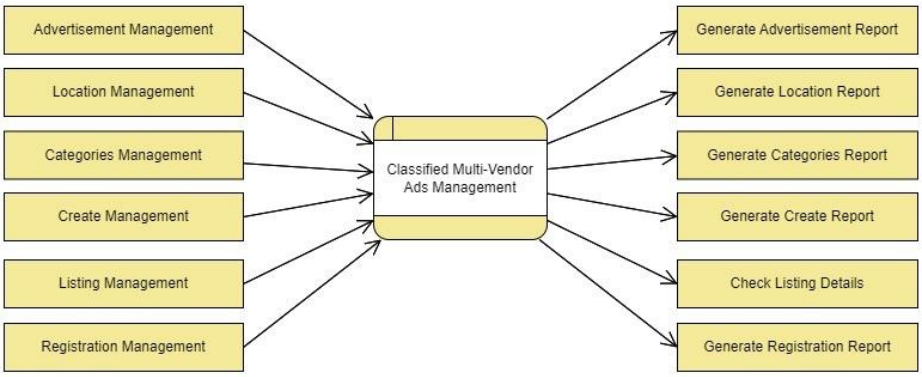
3.9 Data Flow Diagram Analysis

A Data Flow Diagram (DFD) is a graphic illustration that shows how data enters the Classified Multi-Vendor Ads Management Website from outside sources. It is a potent tool for understanding the information system's logical and physical aspects. The DFD offers a graphical representation of the system, including all procedures and components for data storage necessary for system transactions. Employing the DFD allows stakeholders to gain insightful awareness of how data flows through the system and the multiple interactions that occur during various transactions, providing a thorough comprehension of the data flow dynamics of the system.



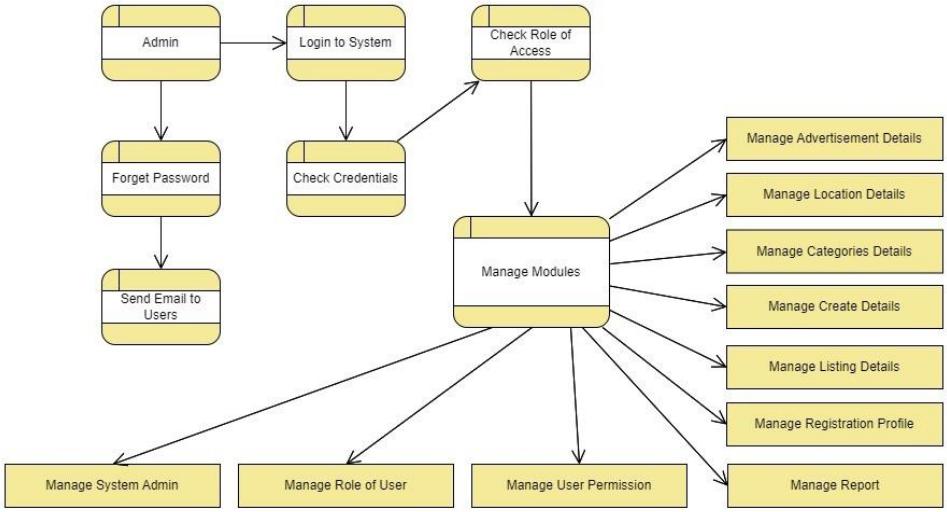
Level 0 DFD of Classified Multi-Vendor Ads Management Website

Figure 3.5 Level 0 DFD of Classified Multi-Vendor Ads Management Website



Level 1 DFD of Classified
Multi-Vendor Ads Management Website

Figure 3.6 Level 1 DFD of Classified Multi-Vendor Ads Management Website



Level 2 DFD of Classified
Multi-Vendor Ads Management Website

Figure 3.7 Level 2 DFD of Classified Multi-Vendor Ads Management Website

Chapter 4 System and Database Design

4.1 User Interface Design Specifications

The Classified Multi-Vendor Ads Management Website incorporates front-end and back-end processes to ensure seamless operations. Our primary objective is to craft an aesthetically pleasing design for the platform while strongly emphasizing user-friendly workflows. I am dedicated to creating a user-friendly management system for both the front-end and back-end components of this web-based Ads Management Website.

Currently, I am actively developing an overview table to offer users a comprehensive view of the system, which includes the following activities:

1. Ads Management: Overseeing and controlling advertising content on the platform.

2. Enhanced System Options: Upgrading the website's features and settings for better performance.

3. User Interface Refinement: Improving the visual and interactive elements of the website to enhance user experience.

4. System Functionality Buttons: Developing buttons on the interface for specific system actions or features.

5. Vendor Management: Handling the accounts and content of sellers using the platform.

6. Vendor Registration: Process for vendors to sign up and create profiles on the website.

7. Listing Management: Organizing and maintaining the product or service listings on the website.

8. User Profiles: Managing creating and maintaining user accounts and their information.

9. User Dashboard: Providing a personalized control panel for users to manage their interactions with the website.

10. API Integration: Incorporating external services and data through Application Programming Interfaces.

11. Listing Moderation: Monitoring and regulating the listings to ensure they meet the platform's standards.

12. Payment Processing: Handling transactions and financial operations on the website.

13. Interactive JavaScript Views: Creating dynamic and responsive user interfaces with JavaScript.

14. Query Optimization: Enhancing the efficiency and speed of database searches and data retrieval.

4.1.1 Interaction Design and UI/UX Principles

The Classified Multi-Vendor Ads Management Website is a user-friendly web-based platform designed to streamline the management of classified advertisements from multiple vendors. This system offers a comprehensive view of all listed ads and ensures user satisfaction remains a top priority. Prioritizing user-friendliness is essential, as complex or cumbersome platforms often drive users to seek alternatives for a more seamless experience. Therefore, maintaining a user-friendly interface is critical for the website's success.

When users find the website easy to navigate and use, they are more likely to actively engage with the platform, leading to an improved overall user experience. This interactive approach significantly boosts user engagement and enhances the website's overall performance. The website's simplicity empowers users to navigate and control the platform effortlessly, further elevating its usability and effectiveness.

4.1.2 Implementation Requirements

1. To design and implement the Classified Multi-Vendor Ads Management Website, I require the following tools and technologies:

- **Integrated Development Environments (IDEs):** Visual Studio Code, Sublime Text
- **Programming Languages and Queries:** HTML, CSS, PHP, Laravel, Bootstrap 4, JavaScript
- **Version Control:** Git and GitHub for collaborative development

2. The website should be compatible with modern web browsers such as Chrome, Firefox, and Edge to ensure a broad reach and consistent user experience.
3. The chosen technology stack includes:
    - **Front-end:** HTML, CSS, JavaScript (JS), Bootstrap 4 for responsive design
    - **Back-end:** PHP and Laravel for server-side scripting and application logic
4. Security measures will be implemented to protect against SQL injection attacks and ensure data integrity and user safety. String escaping and input validation will be critical components of our security strategy.

4.2 Detailed System Design

4.2.1 User System Design

1. Registration

New buyers and sellers can sign up for accounts by providing essential information. The registration process typically includes:

- Full name
- Email address
- Contact number

- Secure password

- Username

- Email verification

- Agree to the terms of service and privacy policy

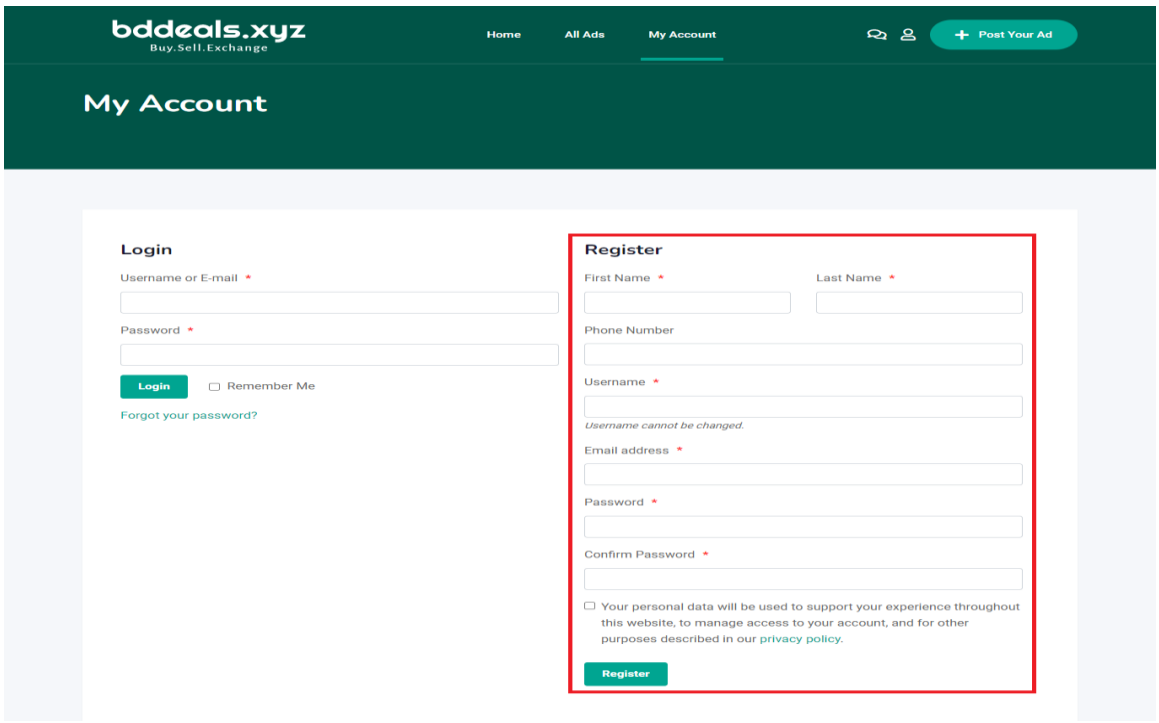


Figure 4.1 User Registration Form

2. Authentication

Login:

- Registered buyers and sellers can both access their accounts by entering the following:

  - Username or email address

  - Password

Password Recovery:

- If buyers or sellers forget their password, they can recover it.

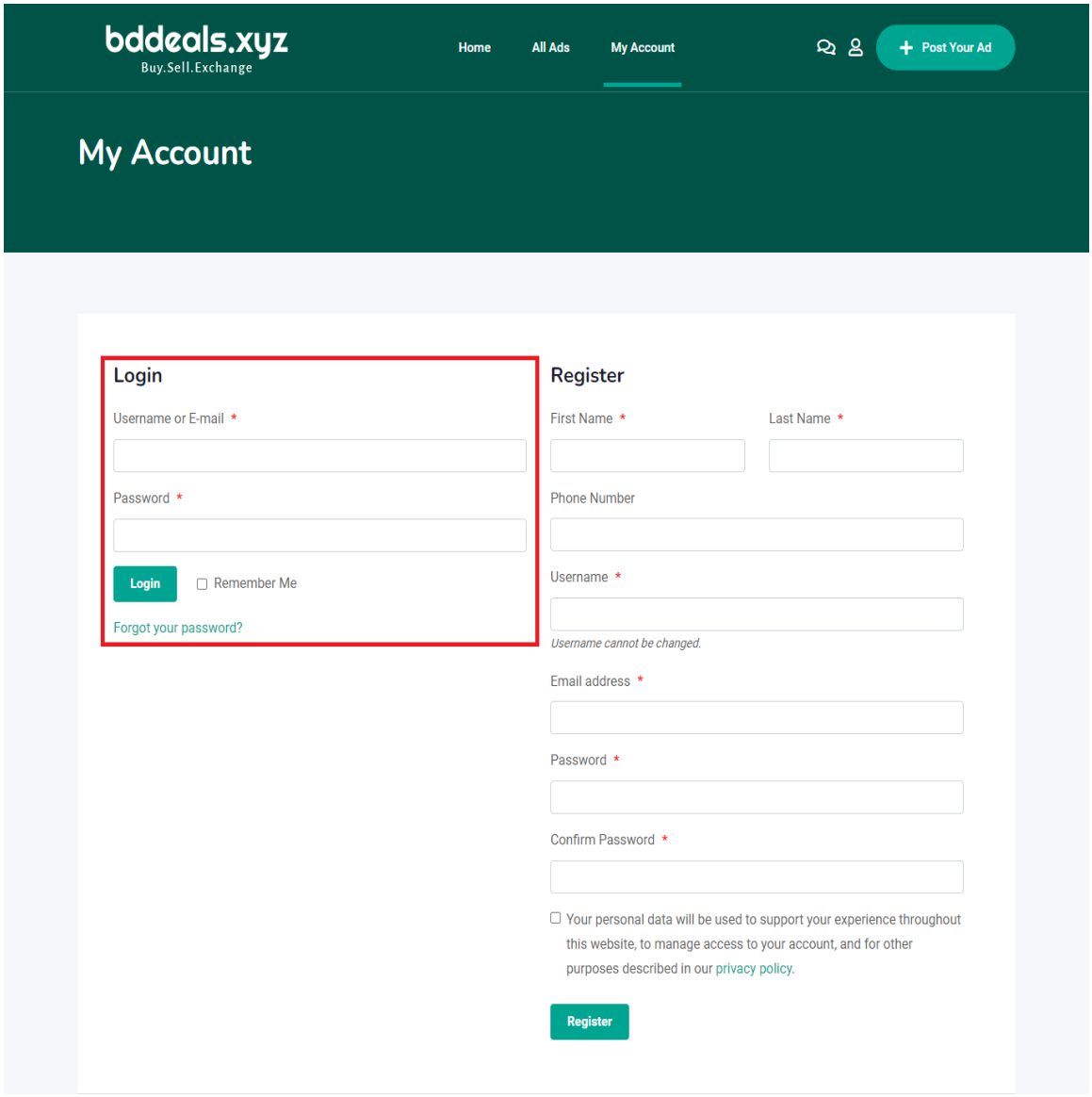


Figure 4.2 User Login Form

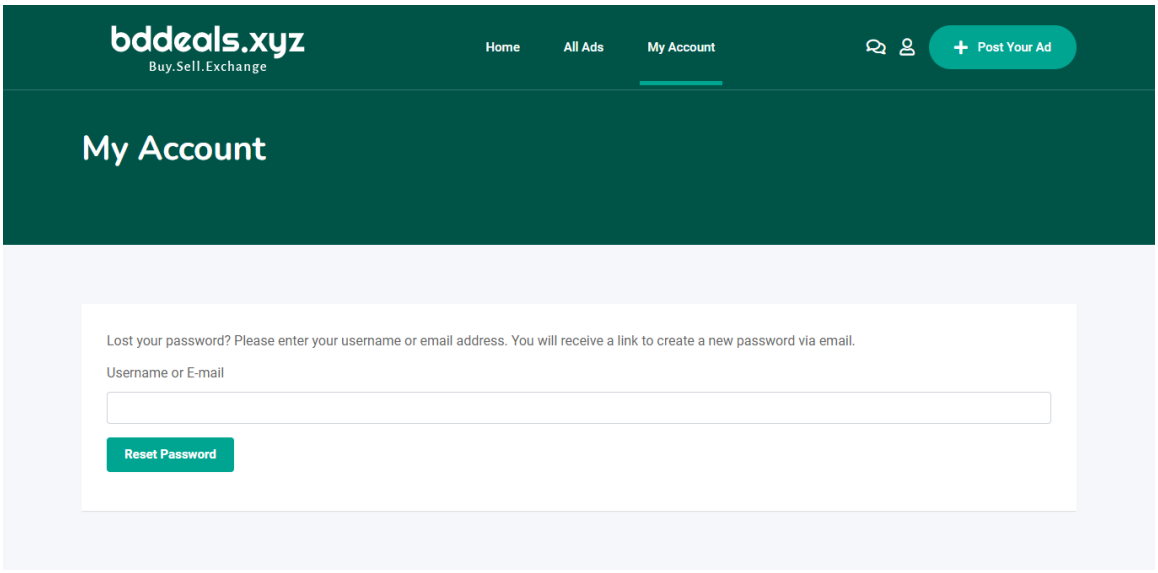


Figure 4.3 Password Reset Page

3. Membership:

Sellers can choose different membership plans based on their needs. Membership features might include:

- Access to additional ad posting features

- Priority placement for listings

- Enhanced visibility in search results

- Subscription and payment management

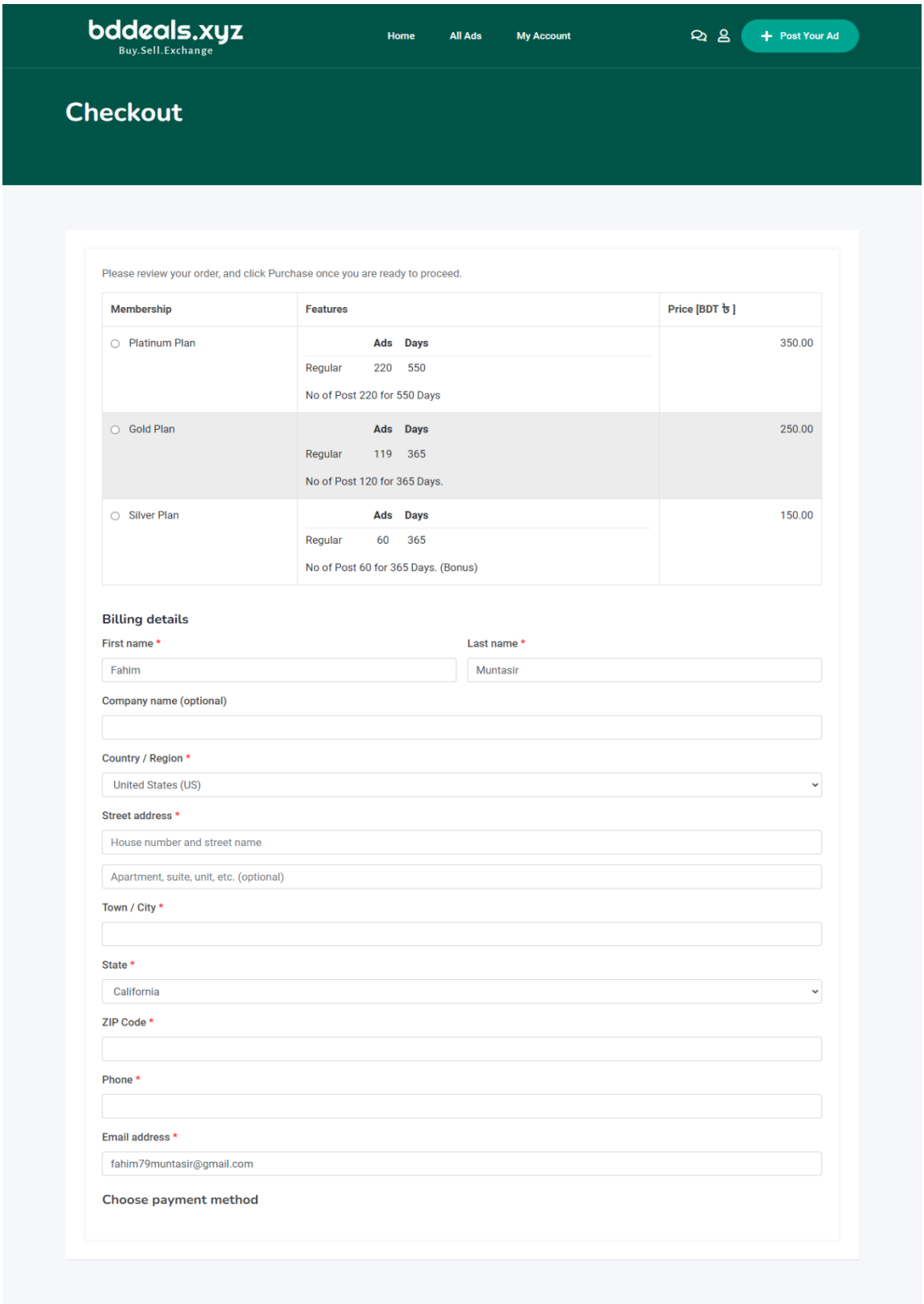


Figure 4.4 Membership Feature Page

4. Ad Search

All users, whether buyers or sellers, can utilize the ad search functionality to find listings that match their interests. Features include:

- Search filters (e.g., category, location, price range)

- Sorting options (e.g., newest, price low to high)

- View details and images, and contact the seller.

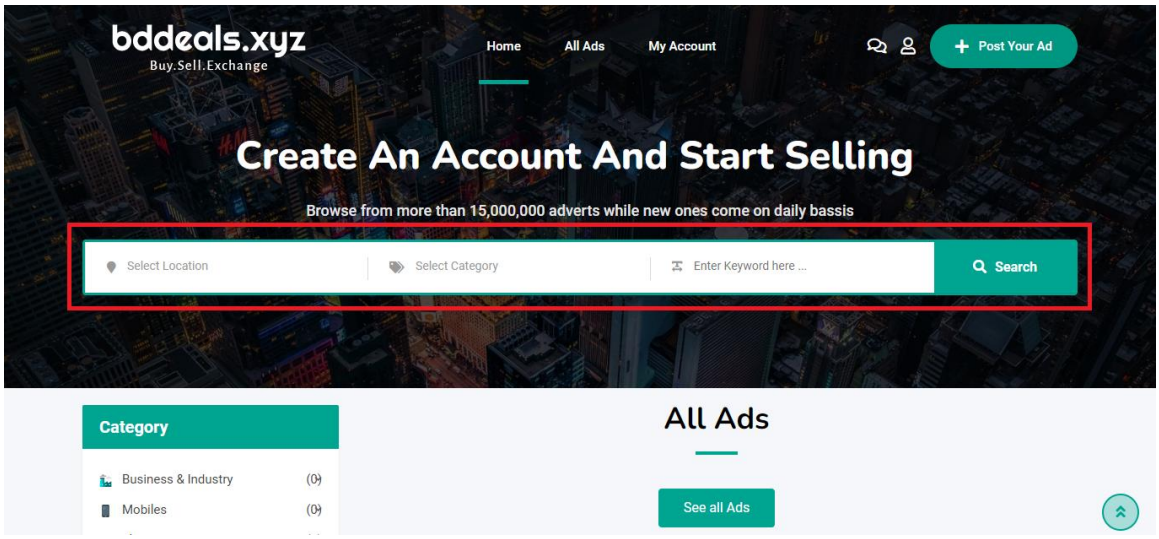


Figure 4.5 Ad Search Option

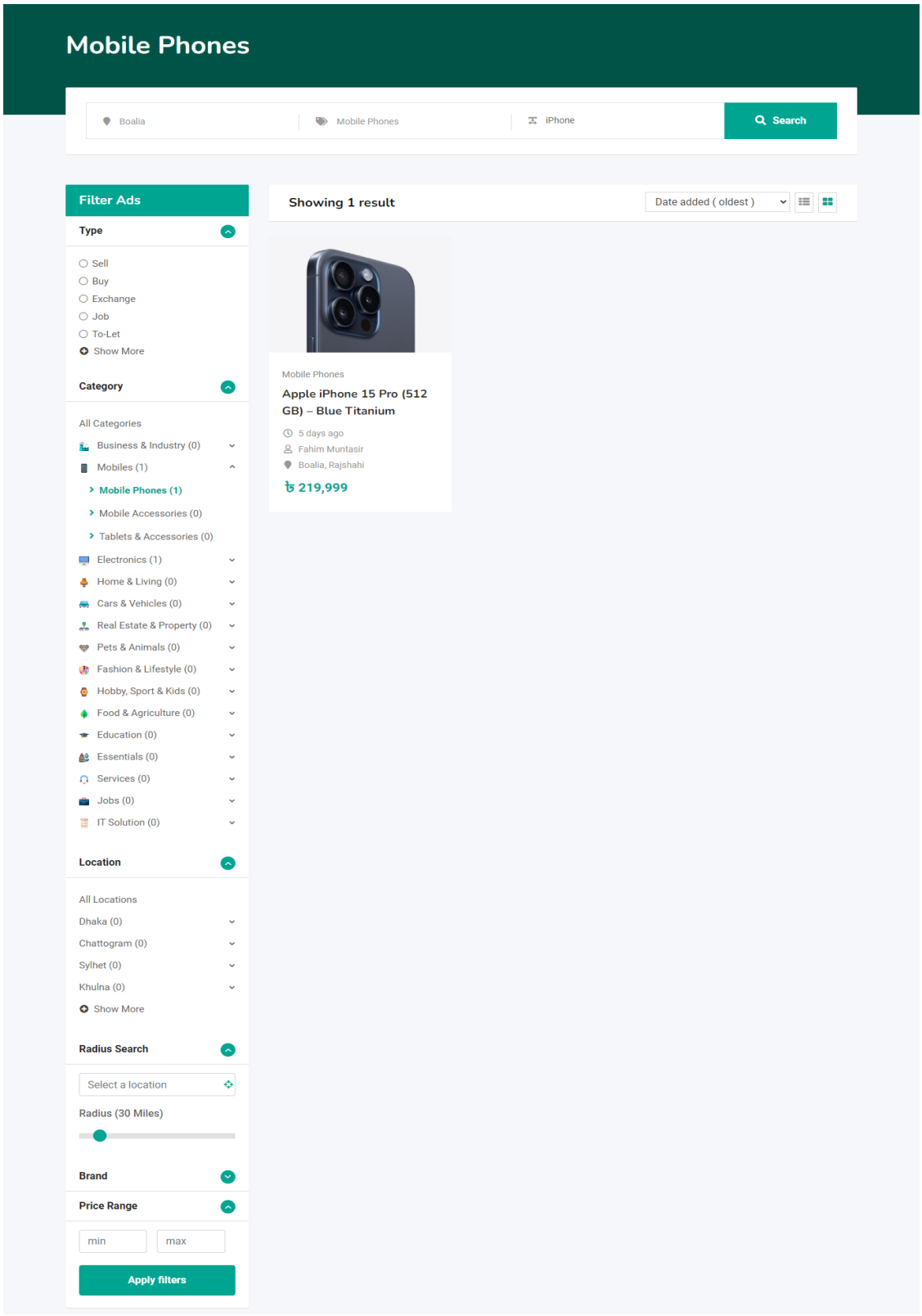


Figure 4.6 Ad Search Result

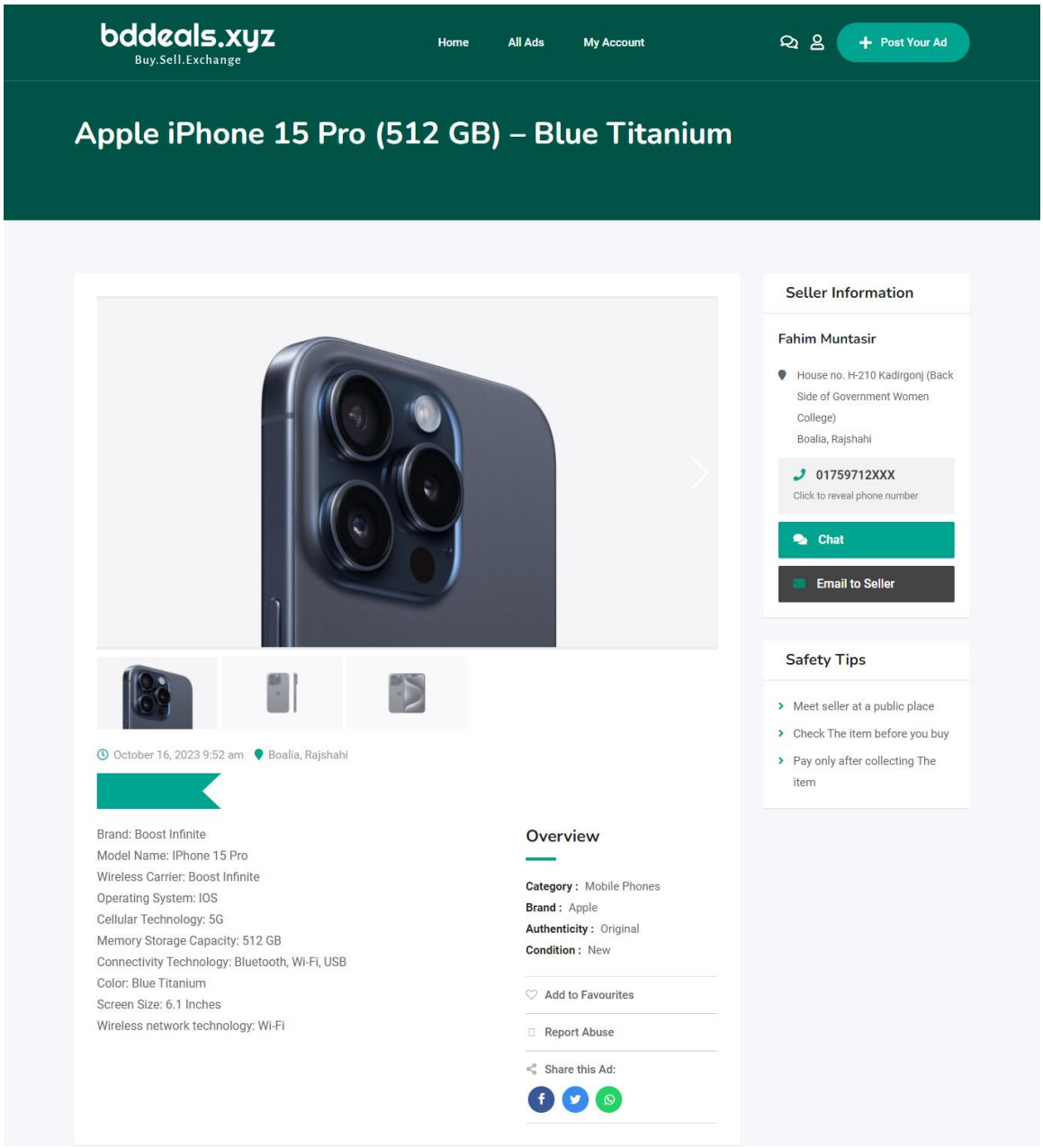


Figure 4.7 Ad Details

5. Ad Post

Seller Ad Posting:

- Sellers can create and publish ads for their products or services with the following steps:

- Title, description, and category selection

- Pricing and negotiation options

- Image uploads

- Contact information

- Review and confirmation before posting.

Ad Management:

- Sellers can edit, renew, or deactivate their ads and track ad performance.

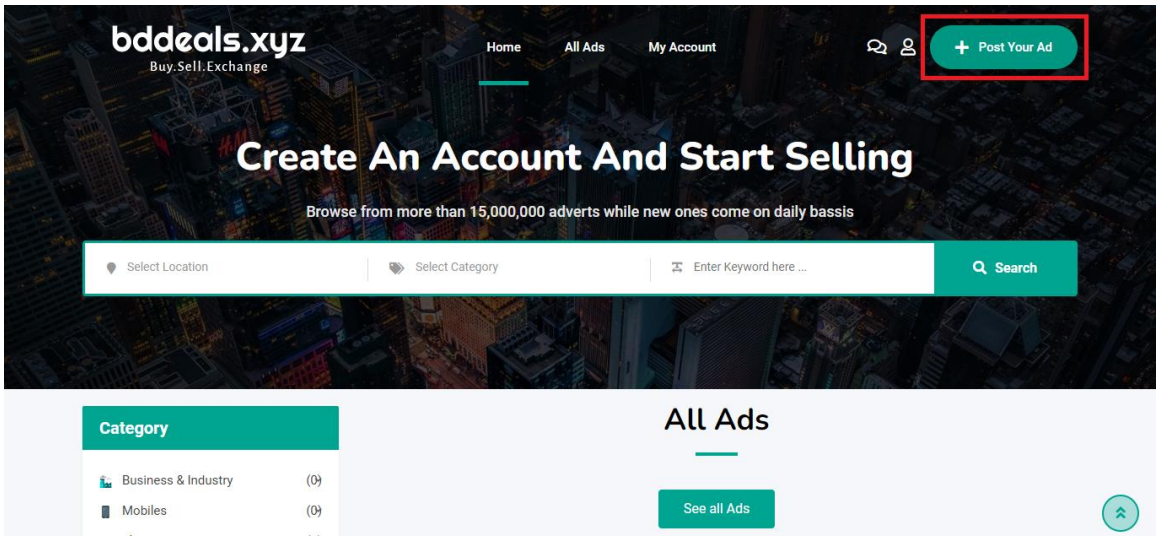


Figure 4.8 Ad Post Option

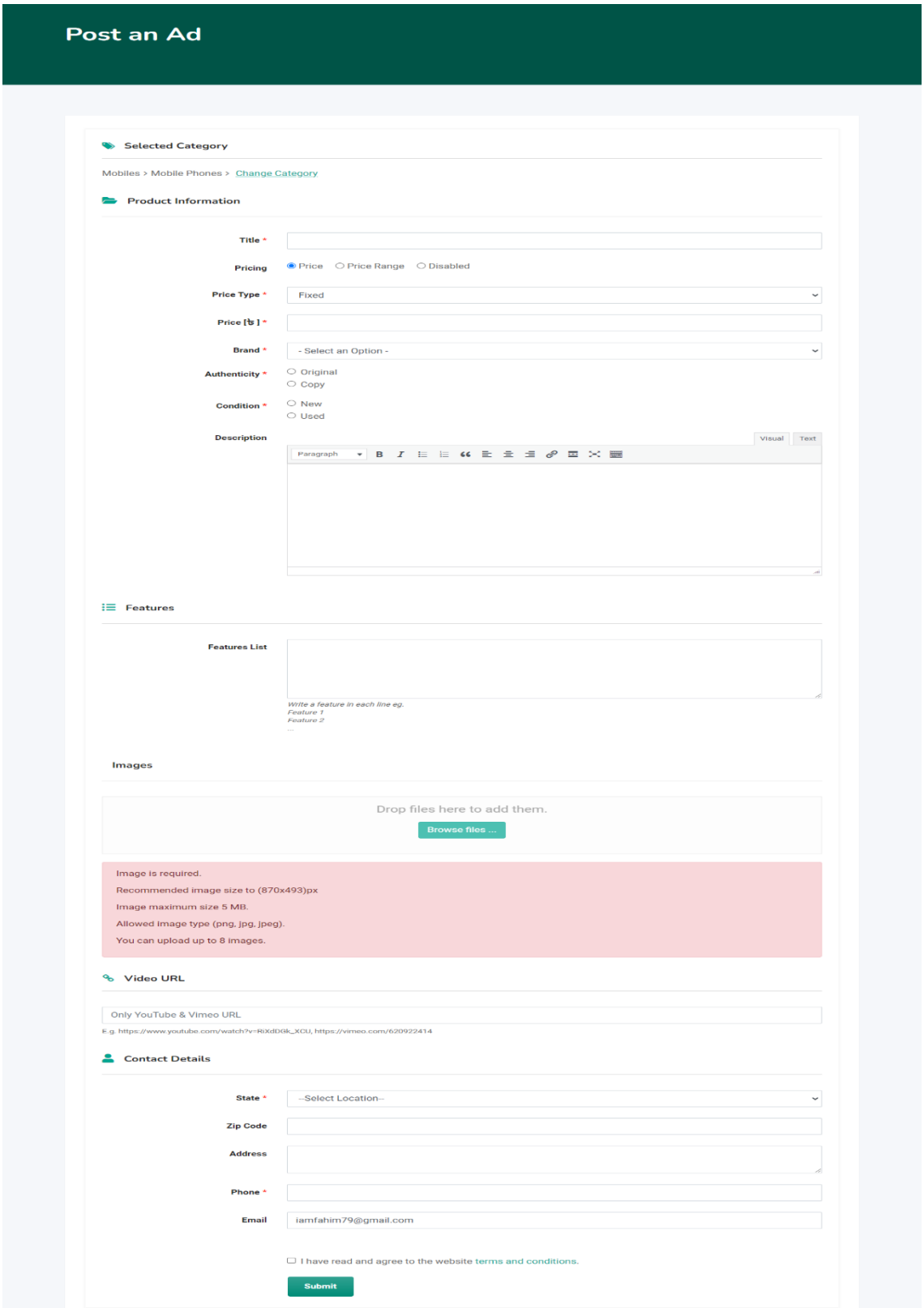


Figure 4.9 Ad Post Page

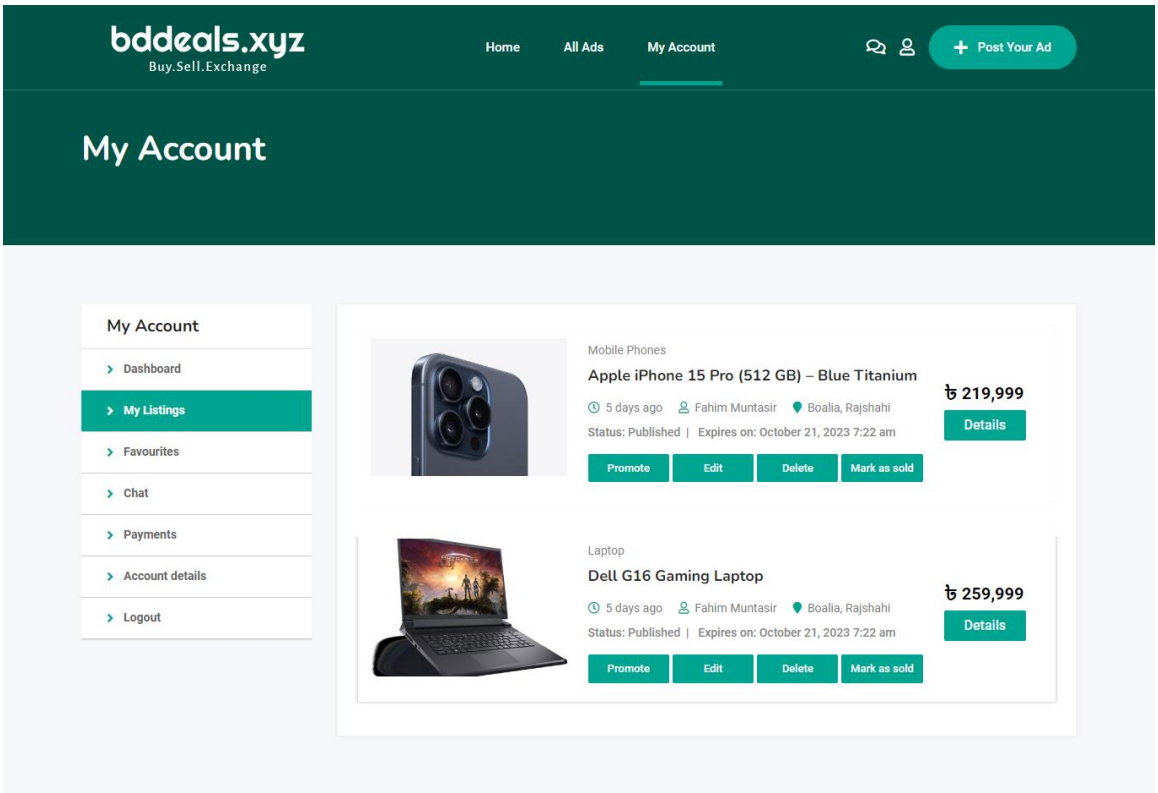


Figure 4.10 Ad Editing, Deleting, Promoting Page

4.2.2 Admin System Design

1. Add Listing, Approve Listing, Listing Deletion

a) Add Listing

- Administrators can add listings directly to the platform, including:
  - Title, description, and category selection
  - Pricing and negotiation options
  - Image uploads
  - Contact information
- This feature allows administrators to assist users with ad creation when needed.

b) Approve Listing

- Listings submitted by users require admin approval before being published. Administrators review and approve listings to ensure they comply with platform guidelines and standards.

c) Listing Deletion

- Admins can remove or deactivate listings that violate platform policies or are reported by users for inappropriate content.

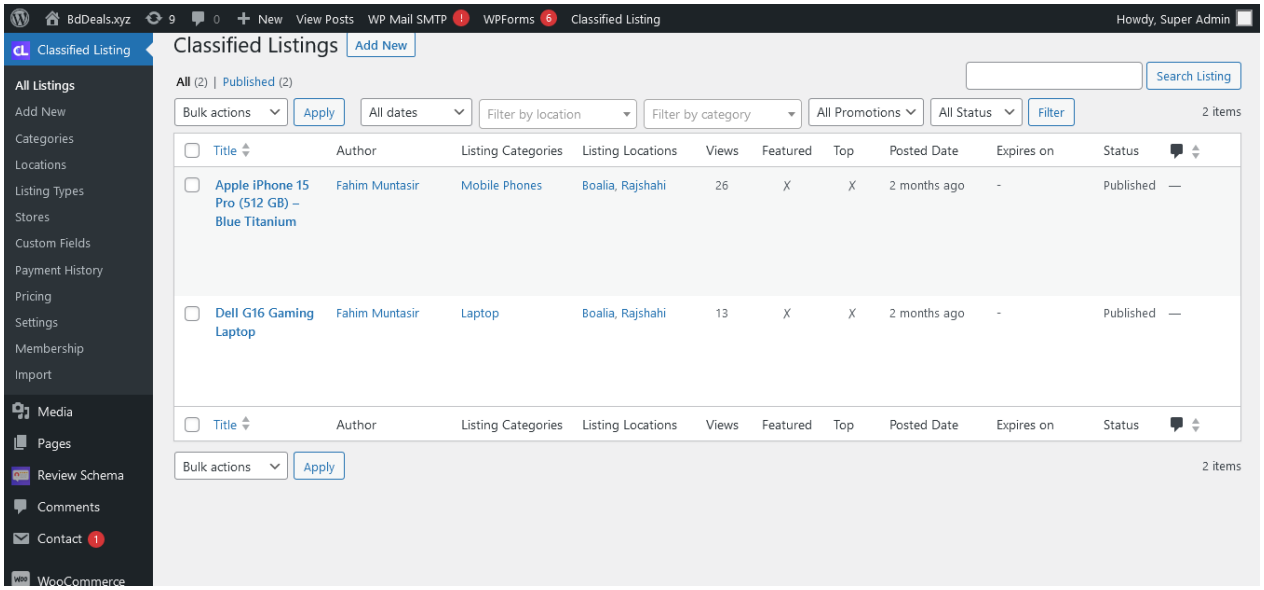


Figure 4.11 Classified Listings

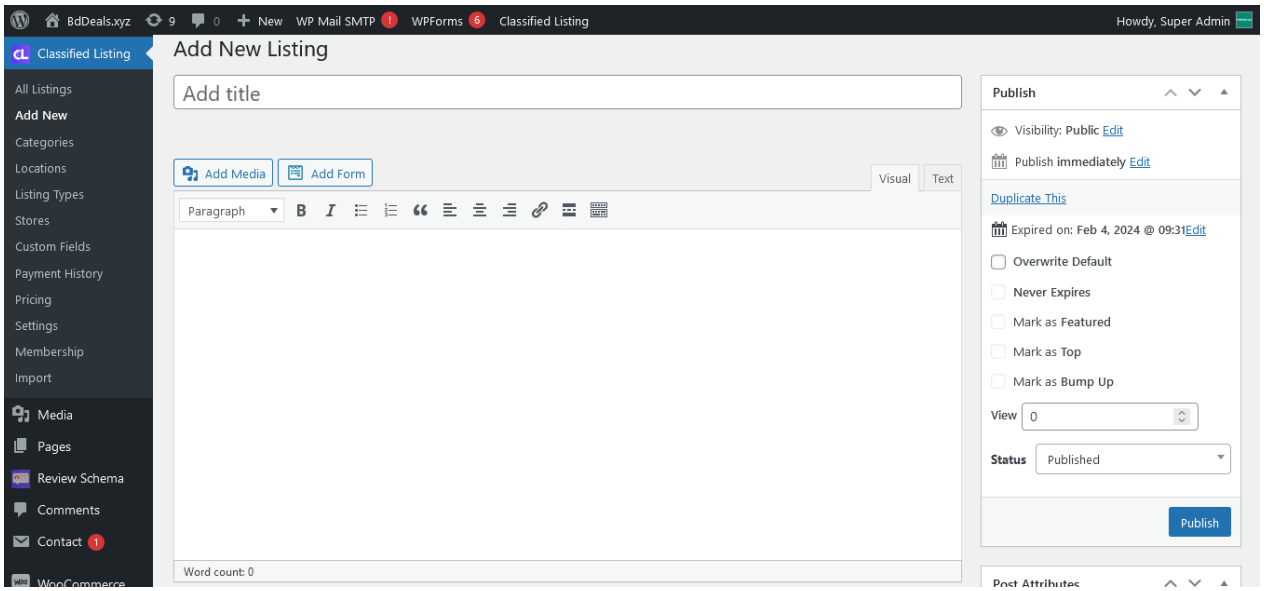


Figure 4.12 Add New Listing

2. Category, List Type, City, and Location Management

- Admins can manage and maintain the platform's categories, list types, cities, and locations. This includes:

- Adding new categories, list types, cities, and locations

- Editing or updating existing ones

- Deactivating or archiving irrelevant or outdated options.

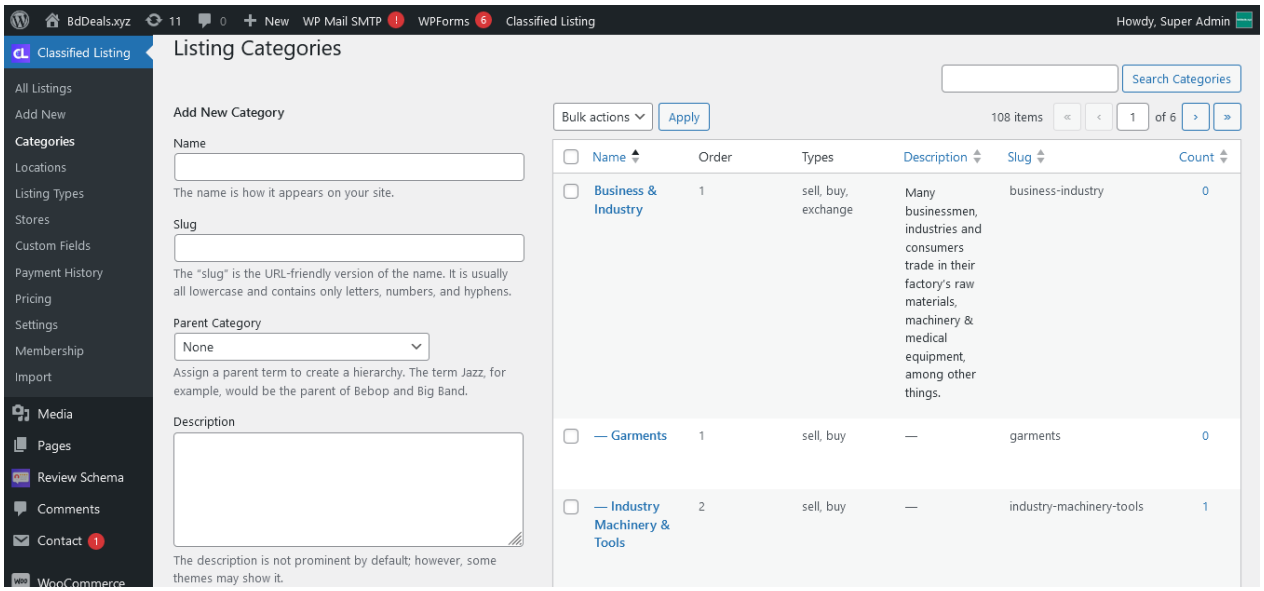


Figure 4.13 Listing Categories

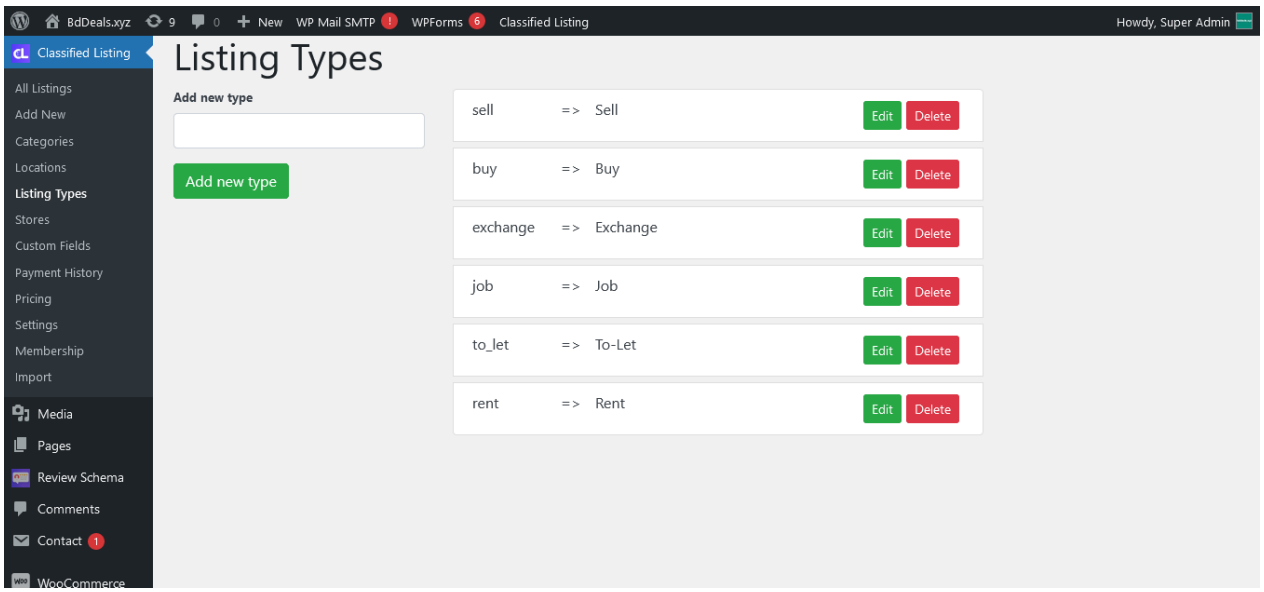


Figure 4.14 Listing Types

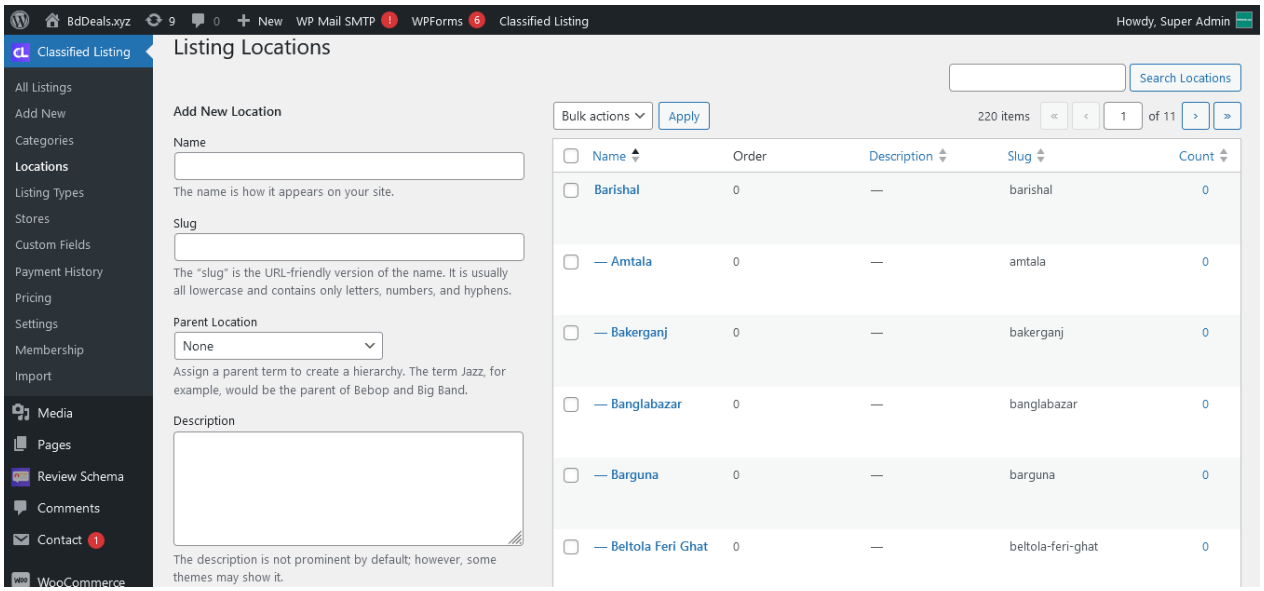


Figure 4.15 Listing Locations

3. Add Membership, Approve Membership, Membership Deletion

a) Add Membership

- Administrators can create and add membership plans with various features and benefits for users. This includes specifying:

- Membership duration

- Pricing

- Associated privileges and benefits.

b) Approve Membership

- Membership requests submitted by users require admin approval. Administrators review and approve membership applications to grant users access to premium features.

c) Membership Deletion

- Administrators can delete user accounts, memberships, and listings when necessary. This may be required in cases of policy violations or user requests.

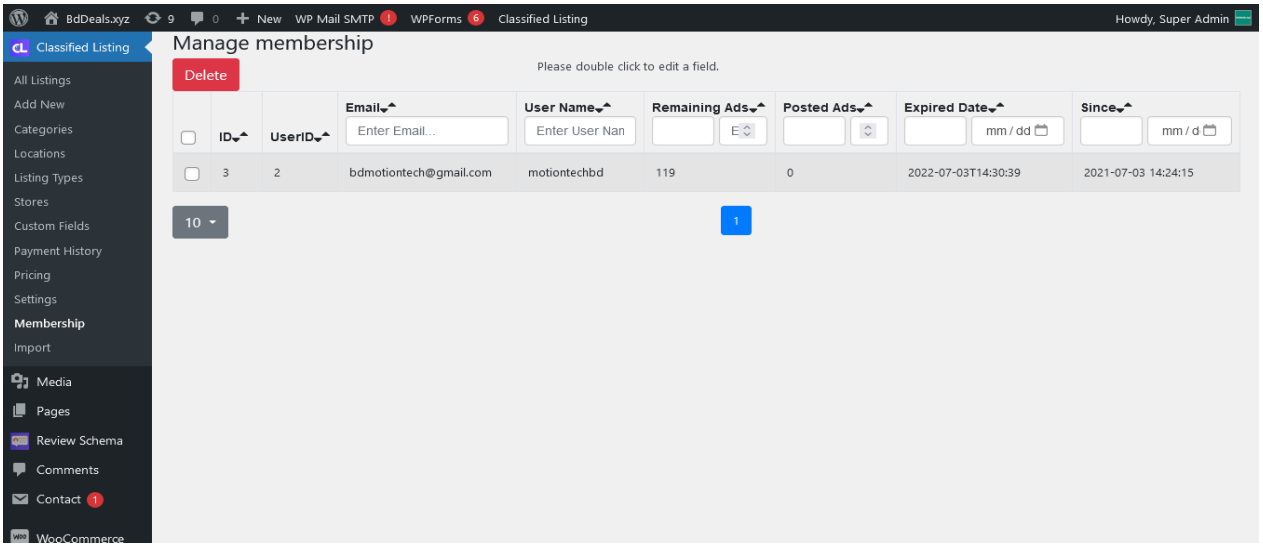


Figure 4.16 Manage Membership

4. Approve Membership Payments

- Admins can review and approve user payments for their selected membership plans. This ensures that users gain the intended benefits of their paid memberships.

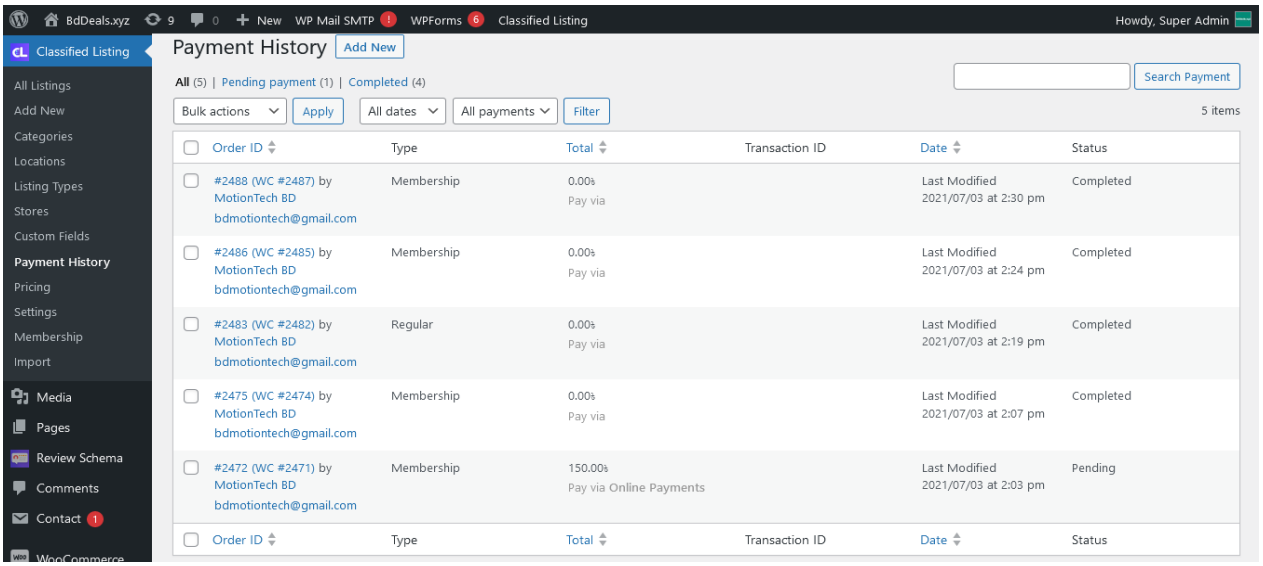


Figure 4.17 Payment History

5. Settings

- Admins can settings so many options like: General, Moderation, Payment, Email, Account & Policy, Style, Misc, Chat, Advanced, Tools, App, Membership and Addon & Theme.

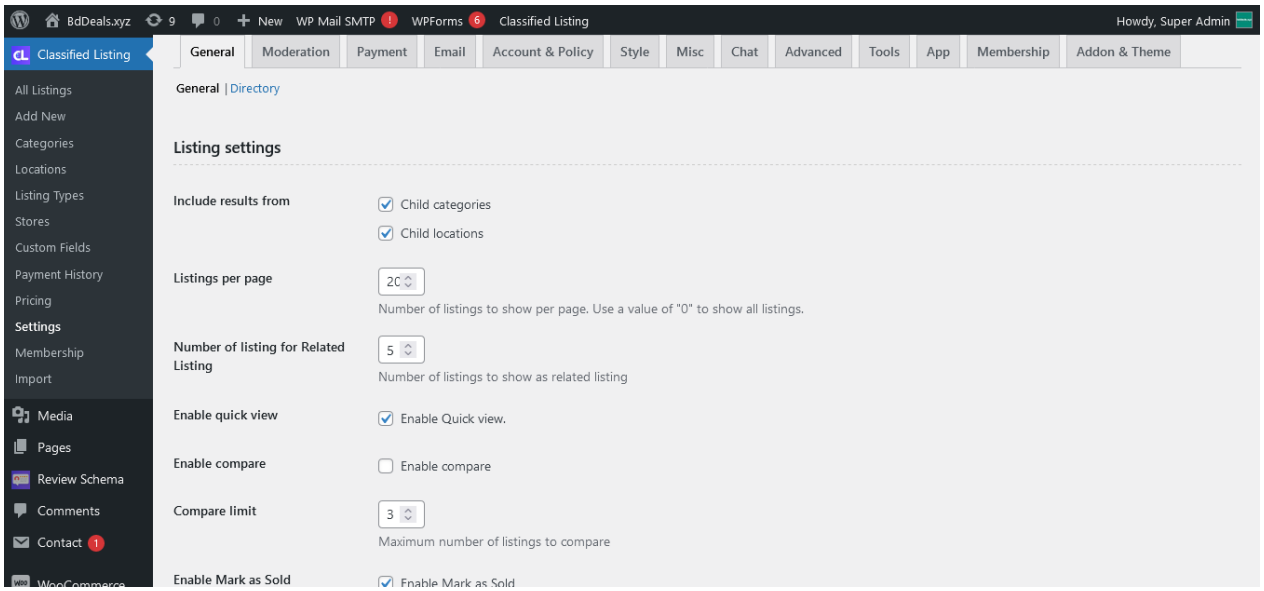


Figure 4.18 Settings Panel

4.3 Implementation of the Database

The database implementation for the Classified Multi-Vendor Ads Management Website involves the meticulous design, setup, and management of a database system to store and retrieve data efficiently. This process typically encompasses the following steps:

- **Database Design:** Begin by identifying the data entities you need to keep and establish how they are related to one another. Create an Entity Relationship (ER) diagram to represent these relationships visually.

- **Choose a Database Management System (DBMS):** Select a DBMS that aligns with the specific requirements of your project. MySQL is often a preferred choice, especially when working with PHP, owing to its robustness and ease of use.

- **Database Setup:** Set up a new database within your chosen DBMS. If you use MySQL,

you can accomplish this through tools like phpMyAdmin or the MySQL command line.

- **Table Creation:** Create tables in your database based on the entities outlined in your ER diagram. Ensure you choose appropriate data types for each column to represent the data accurately.

- **Establish Relationships:** Implement essential relationships between tables, including defining foreign keys to effectively handle one-to-many or many-to-many relationships.

- **Initial Data Input:** Populate the database with any initial data required for your system to function correctly. This may include default admin user accounts, vendor details, categories, or initial listings.

- **Query Development:** Craft SQL queries to perform essential database operations such as creating, reading, updating, and deleting data. These queries will be utilized in your PHP code to interact with the database.

- **Database Optimization:** Enhance database performance by indexing frequently queried columns for faster retrieval. Normalize your data to eliminate redundancy and maintain data integrity through proper constraints.

- **Database Testing:** Rigorously test your database with your application to ensure it operates correctly and efficiently. Verify that all CRUD (Create, Read, Update, Delete) operations function as expected across all tables.

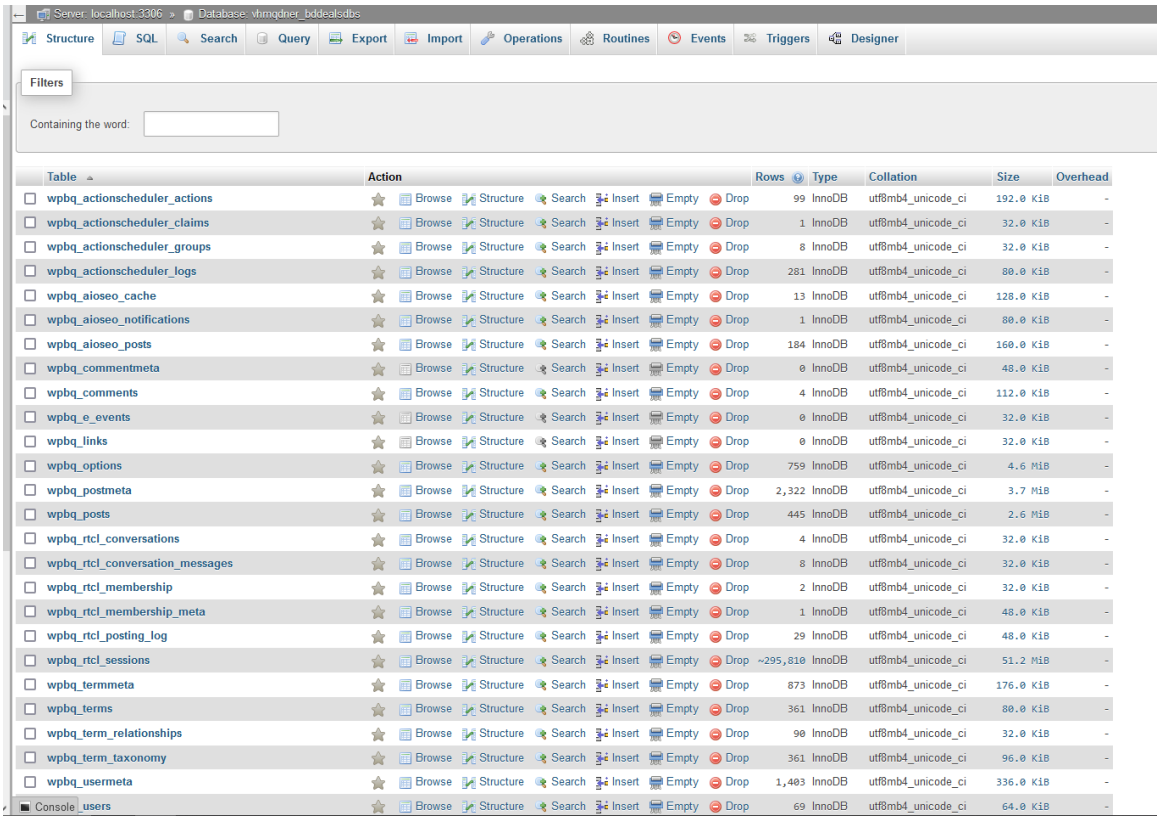
- **Backup and Recovery:** Implement a robust strategy for regular database backups and establish a recovery plan in case of data loss, ensuring the security and availability of your critical data.

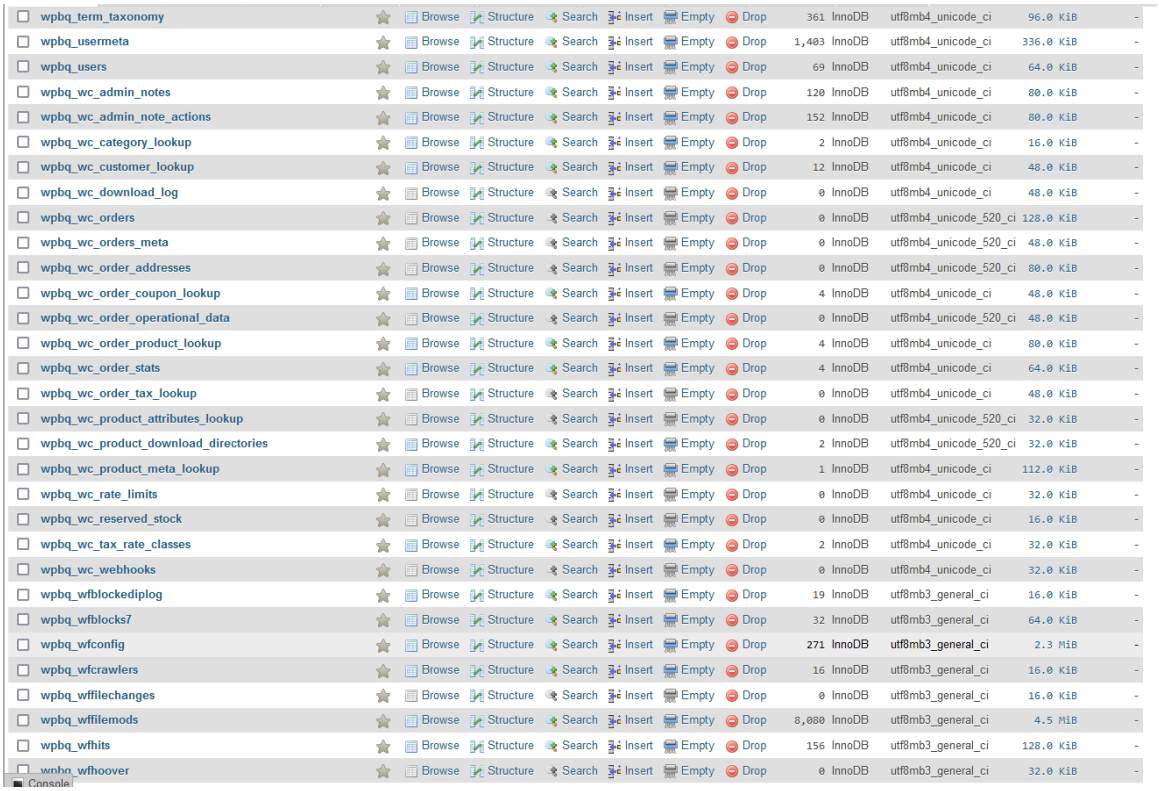
4.4 Database Schema and Design

The database is the backbone of the Classified Multi-Vendor Ads Management Website, storing and managing all the content and user data. A MySQL database has been installed and

meticulously configured to ensure data integrity and provide the necessary functionality for the application to function seamlessly.

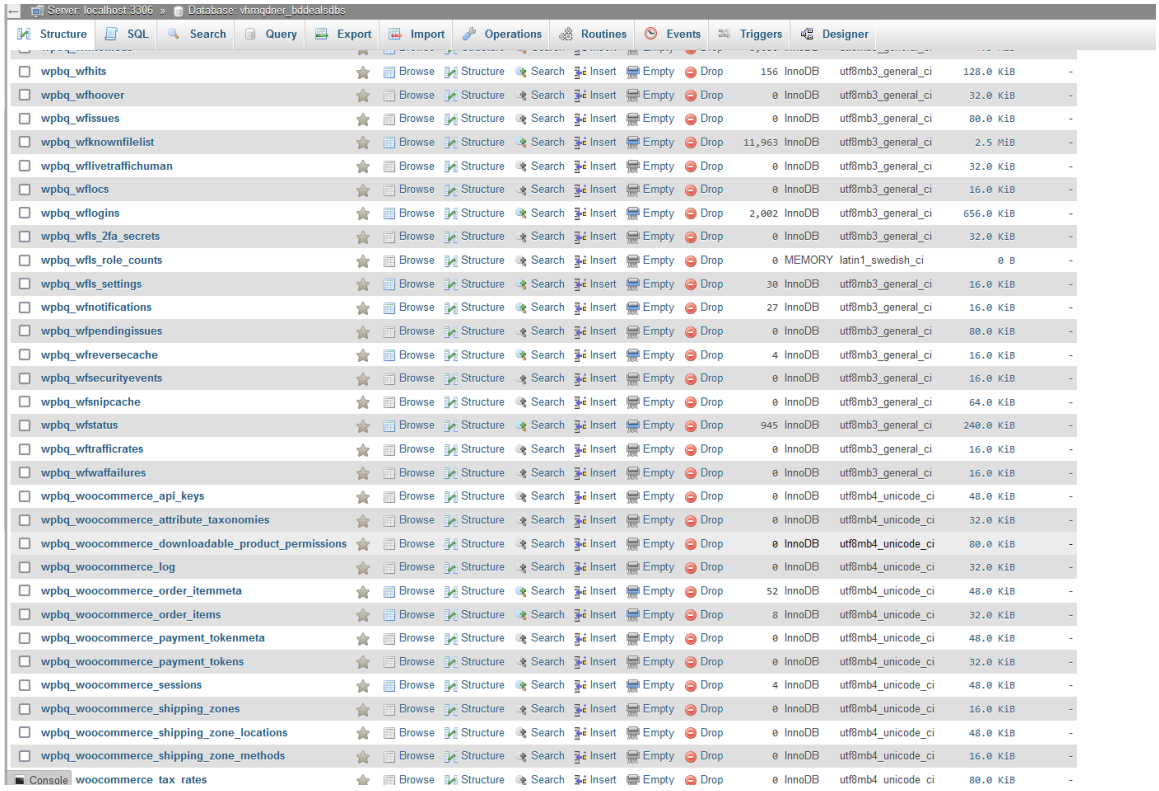
1. Main Database: Here is the entire database of this project.





wpbq_term_taxonomy	Browse	Structure	Search	Insert	Empty	Drop	361	InnoDB	utf8mb4_unicode_ci	96.0 KiB
wpbq_usermeta	Browse	Structure	Search	Insert	Empty	Drop	1,403	InnoDB	utf8mb4_unicode_ci	336.0 KiB
wpbq_users	Browse	Structure	Search	Insert	Empty	Drop	69	InnoDB	utf8mb4_unicode_ci	64.0 KiB
wpbq_wc_admin_notes	Browse	Structure	Search	Insert	Empty	Drop	120	InnoDB	utf8mb4_unicode_ci	80.0 KiB
wpbq_wc_admin_note_actions	Browse	Structure	Search	Insert	Empty	Drop	152	InnoDB	utf8mb4_unicode_ci	80.0 KiB
wpbq_wc_category_lookup	Browse	Structure	Search	Insert	Empty	Drop	2	InnoDB	utf8mb4_unicode_ci	16.0 KiB
wpbq_wc_customer_lookup	Browse	Structure	Search	Insert	Empty	Drop	12	InnoDB	utf8mb4_unicode_ci	48.0 KiB
wpbq_wc_download_log	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB
wpbq_wc_orders	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_520_ci	128.0 KiB
wpbq_wc_orders_meta	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_520_ci	48.0 KiB
wpbq_wc_order_addresses	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_520_ci	80.0 KiB
wpbq_wc_order_coupon_lookup	Browse	Structure	Search	Insert	Empty	Drop	4	InnoDB	utf8mb4_unicode_ci	48.0 KiB
wpbq_wc_order_operational_data	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_520_ci	48.0 KiB
wpbq_wc_order_product_lookup	Browse	Structure	Search	Insert	Empty	Drop	4	InnoDB	utf8mb4_unicode_ci	80.0 KiB
wpbq_wc_order_stats	Browse	Structure	Search	Insert	Empty	Drop	4	InnoDB	utf8mb4_unicode_ci	64.0 KiB
wpbq_wc_order_tax_lookup	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB
wpbq_wc_product_attributes_lookup	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_520_ci	32.0 KiB
wpbq_wc_product_download_directories	Browse	Structure	Search	Insert	Empty	Drop	2	InnoDB	utf8mb4_unicode_520_ci	32.0 KiB
wpbq_wc_product_meta_lookup	Browse	Structure	Search	Insert	Empty	Drop	1	InnoDB	utf8mb4_unicode_ci	112.0 KiB
wpbq_wc_rate_limits	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB
wpbq_wc_reserved_stock	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 KiB
wpbq_wc_tax_rate_classes	Browse	Structure	Search	Insert	Empty	Drop	2	InnoDB	utf8mb4_unicode_ci	32.0 KiB
wpbq_wc_webhooks	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB
wpbq_wfblockediplog	Browse	Structure	Search	Insert	Empty	Drop	19	InnoDB	utf8mb3_general_ci	16.0 KiB
wpbq_wfblocks7	Browse	Structure	Search	Insert	Empty	Drop	32	InnoDB	utf8mb3_general_ci	64.0 KiB
wpbq_wfconfig	Browse	Structure	Search	Insert	Empty	Drop	271	InnoDB	utf8mb3_general_ci	2.3 MiB
wpbq_wfcrawlers	Browse	Structure	Search	Insert	Empty	Drop	16	InnoDB	utf8mb3_general_ci	16.0 KiB
wpbq_wffilechanges	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb3_general_ci	16.0 KiB
wpbq_wffilemods	Browse	Structure	Search	Insert	Empty	Drop	8,080	InnoDB	utf8mb3_general_ci	4.5 MiB
wpbq_wfhits	Browse	Structure	Search	Insert	Empty	Drop	156	InnoDB	utf8mb3_general_ci	128.0 KiB
wpbq_wfhoover	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb3_general_ci	32.0 KiB

Console



Server: localhost:3306 » Database: vhmqdner_bddealsdbs

Structure | SQL | Search | Query | Export | Import | Operations | Routines | Events | Triggers | Designer

wpbq_wfhits	Browse	Structure	Search	Insert	Empty	Drop	156	InnoDB	utf8mb3_general_ci	128.0 KiB
wpbq_wfhoover	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb3_general_ci	32.0 KiB
wpbq_wfissues	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb3_general_ci	80.0 KiB
wpbq_wfknownfilelist	Browse	Structure	Search	Insert	Empty	Drop	11,963	InnoDB	utf8mb3_general_ci	2.5 MiB
wpbq_wflivetraffichuman	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb3_general_ci	32.0 KiB
wpbq_wflocs	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb3_general_ci	16.0 KiB
wpbq_wflogins	Browse	Structure	Search	Insert	Empty	Drop	2,002	InnoDB	utf8mb3_general_ci	656.0 KiB
wpbq_wfls_2fa_secrets	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb3_general_ci	32.0 KiB
wpbq_wfls_role_counts	Browse	Structure	Search	Insert	Empty	Drop	0	MEMORY	latin1_swedish_ci	0 B
wpbq_wfls_settings	Browse	Structure	Search	Insert	Empty	Drop	30	InnoDB	utf8mb3_general_ci	16.0 KiB
wpbq_wfnotifications	Browse	Structure	Search	Insert	Empty	Drop	27	InnoDB	utf8mb3_general_ci	16.0 KiB
wpbq_wfpendingissues	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb3_general_ci	80.0 KiB
wpbq_wfreversecache	Browse	Structure	Search	Insert	Empty	Drop	4	InnoDB	utf8mb3_general_ci	16.0 KiB
wpbq_wfsecurityevents	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb3_general_ci	16.0 KiB
wpbq_wfsnipcache	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb3_general_ci	64.0 KiB
wpbq_wfstatus	Browse	Structure	Search	Insert	Empty	Drop	945	InnoDB	utf8mb3_general_ci	240.0 KiB
wpbq_wftrafficrates	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb3_general_ci	16.0 KiB
wpbq_wfwaffailures	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb3_general_ci	16.0 KiB
wpbq_woocommerce_api_keys	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB
wpbq_woocommerce_attribute_taxonomies	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB
wpbq_woocommerce_downloadable_product_permissions	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	80.0 KiB
wpbq_woocommerce_log	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB
wpbq_woocommerce_order_itemmeta	Browse	Structure	Search	Insert	Empty	Drop	52	InnoDB	utf8mb4_unicode_ci	48.0 KiB
wpbq_woocommerce_order_items	Browse	Structure	Search	Insert	Empty	Drop	8	InnoDB	utf8mb4_unicode_ci	32.0 KiB
wpbq_woocommerce_payment_tokenmeta	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB
wpbq_woocommerce_payment_tokens	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB
wpbq_woocommerce_sessions	Browse	Structure	Search	Insert	Empty	Drop	4	InnoDB	utf8mb4_unicode_ci	48.0 KiB
wpbq_woocommerce_shipping_zones	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 KiB
wpbq_woocommerce_shipping_zone_locations	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB
wpbq_woocommerce_shipping_zone_methods	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 KiB
Console woocommerce_tax_rates	Browse	Structure	Search	Insert	Empty	Drop	0	InnoDB	utf8mb4_unicode_ci	80.0 KiB

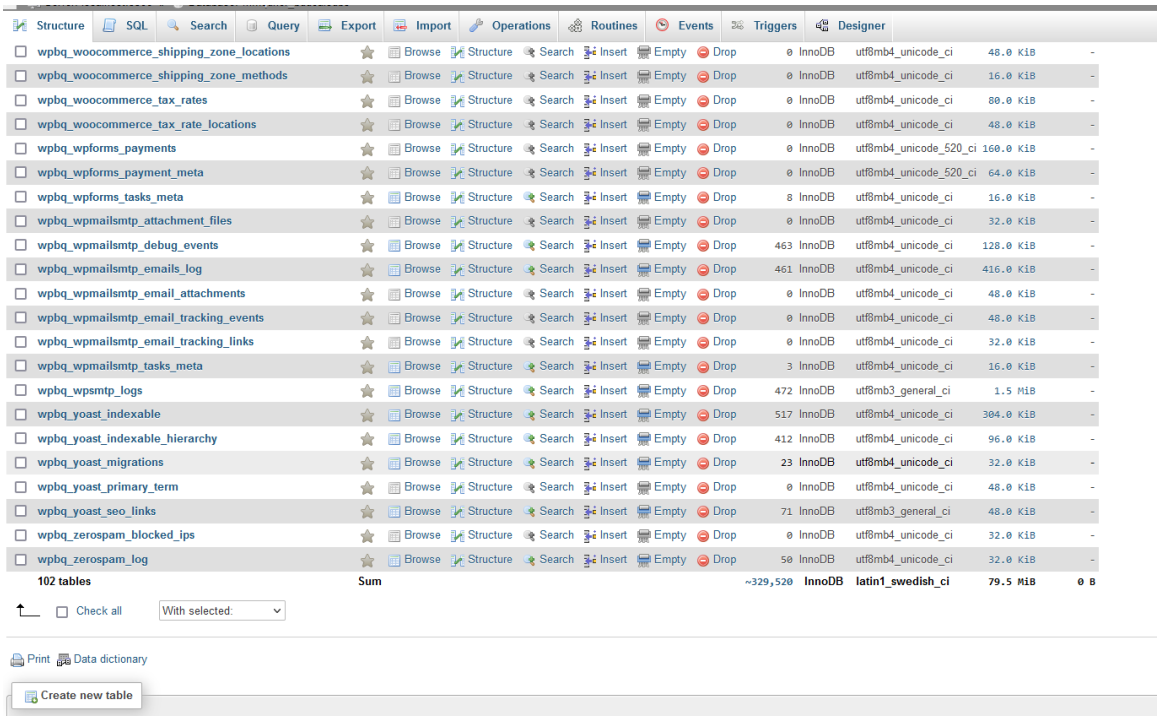


Figure 4.11 Main Database of CMVAMW

2. User Table: This table comprises ten fields with data types, including bigint, varchar, datetime, and int. The 'ID' serves as the primary key. Other fields are user_login, user_pass, user_nickname, user_email, user_url, user_registered, user_activation_key, user_status, and display_name. These fields collectively handle user credentials, personal identification, contact details, registration information, account activation, status indicators, and the display name for user profiles within a system.

No.	Field Name	Data Type	Null	Description
1	ID	bright	No	User's unique ID
2	user_login	varchar(60)	No	Login name
3	user_pass	varchar(255)	No	Password
4	user_nickname	varchar(50)	No	Nickname
5	user_email	varchar(100)	No	Email address
6	user_url	varchar(100)	No	Website link

7	user_registered	datetime	No	Registration date
8	user_activation_key	varchar(255)	No	Activation code
9	user_status	int	No	Account status
10	display_name	varchar(250)	No	Visible name

Table 4.1 User Database Table.

3. Login Table: This table contains nine fields designed to capture various aspects of logging data. The fields include 'id' of type bigint, 'hitID' and 'userID' of type int, 'ctime' as double(17,6), 'fail' as a tinyint, 'action' and 'username' as varchar with lengths of 40 and 255 respectively, 'IP' as binary(16), and 'UA' as text. The 'id' field is likely the primary key. This structure is adept at recording log details such as timestamps ('ctime'), user actions ('action,' 'username'), success or failure states ('fail'), and technical information like IP addresses and user agents ('IP,' 'UA').

No.	Field Name	Data Type	Null	Description
1	id	bigint	No	Main ID
2	hitID	int	Yes	Secondary ID
3	ctime	double(17,6)	No	Time
4	fail	tinyint	No	Error flag
5	action	varchar(40)	No	Action type
6	username	varchar(255)	No	Login name
7	userID	int	No	User ID
8	IP	binary(16)	Yes	IP address
9	UA	text	Yes	Browser info

Table 4.2 Login Database Table.

4. Membership Table: This table consists of 7 fields with data types bigint, int, datetime, and tinyint(1). The primary key of the table is the 'id' field. Other fields include user_id, ads, posted_ads, expiry_date, member_since, and active. These fields track user advertising activities, such as the number of ads a user has, the number of ads posted by a user, the expiry dates of these

ads, the duration of the user's membership, and a boolean indicator (represented as tinyint) of whether the user's ads are currently active.

No.	Field Name	Data Type	Null	Description
1	id	bigint	No	Unique ID
2	user_id	int	No	User ID
3	ads	int	No	Ad count
4	posted_ads	int	No	Ads posted
5	expiry_date	datetime	No	Ad expiry
6	member_since	datetime	No	Start date
7	active	tinyint(1)	No	Account status

Table 4.3 Membership Database Table.

5. Category Table: The table comprises two fields with data types bigint. The fields are 'category_tree_id' and 'category_id,' designed to hold integer values. No explicit primary key is defined, or auto-increment is set in the provided structure. These fields are likely used to manage hierarchical relationships within categories, where 'category_tree_id' could reference the parent category and 'category_id' the identifier for individual types within that tree.

No.	Field Name	Data Type	Null	Description
1	category_tree_id	bigint	No	Main category ID
2	category_id	bigint	No	Subcategory ID

Table 4.4 Category Database Table.

6. Posts Table: This table encompasses 23 fields with various data types such as bigint, varchar, text, datetime, and int. The 'ID' field is the primary key and has been set to auto-increment. The table includes fields such as post_author, post_date, post_date_gmt, post_content, post_title, post_excerpt, post_status, comment_status, ping_status, post_password, post_name, to_ping, pinged, post_modified, post_modified_gmt, post_content_filtered, post_parent, guid, menu_order, post_type, post_mime_type, and comment_count. In a content management system, these fields

collectively manage the metadata and content of posts, such as authorship, publication status, timings, and relationships between posts.

No.	Field Name	Data Type	Null	Description
1	ID	bigint	No	Unique post ID
2	post_author	bigint	No	Author's ID
3	post_date	datetime	No	Creation date
4	post_date_gmt	datetime	No	GMT creation date
5	post_content	longtext	No	Post's text
6	post_title	text	No	Title of the post
7	post_excerpt	text	No	Post summary
8	post_status	varchar(20)	No	Status of post
9	comment_status	varchar(20)	No	Comments status
10	ping_status	varchar(20)	No	Pings status
11	post_password	varchar(255)	No	Post's password
12	post_name	varchar(200)	No	Post's slug
13	to_ping	text	No	URLs to ping
14	pinged	text	No	Pinged URLs
15	post_modified	datetime	No	Last modified date
16	post_modified_gmt	datetime	No	GMT modified date
17	post_content_filtered	longtext	No	Processed content
18	post_parent	bigint	No	Parent post ID
19	guid	varchar(255)	No	Global unique identifier

20	menu_order	int	No	Navigation order
21	post_type	varchar(20)	No	Type of post
22	post_mime_type	varchar(100)	No	MIME type
23	comment_count	bigint	No	Number of comments

Table 4.5 Posts Database Table.

7. Order Table: This table is structured with 17 fields, incorporating data types such as bigint, varchar, decimal, datetime, and text. The 'id' field is the primary key with an auto-increment setting. The table comprises areas including status, currency, type, tax_amount, total_amount, customer_id, billing_email, date_created_gmt, date_updated_gmt, parent_order_id, payment_method, payment_method_title, transaction_id, ip_address, user_agent, and customer_note. These fields are designed to store detailed information regarding orders, such as transaction details, customer information, financial calculations, timestamps, and various identifiers pertinent to e-commerce operations.

No.	Field Name	Data Type	Null	Description
1	id	bigint	No	Unique ID
2	status	varchar(20)	Yes	Order status
3	currency	varchar(10)	Yes	Currency code
4	type	varchar(20)	Yes	Transaction type
5	tax_amount	decimal(26,8)	Yes	Tax value
6	total_amount	decimal(26,8)	Yes	Total value
7	customer_id	bigint	Yes	Customer ID
8	billing_email	varchar(320)	Yes	Billing email
9	date_created_gmt	datetime	Yes	GMT creation time
10	date_updated_gmt	datetime	Yes	GMT update time

11	parent_order_id	bigint	Yes	Parent order ID
12	payment_method	varchar(100)	Yes	Payment method
13	payment_method_title	text	Yes	Payment title
14	transaction_id	varchar(100)	Yes	Transaction ID
15	ip_address	varchar(100)	Yes	IP address
16	user_agent	text	Yes	Browser used
17	customer_note	text	Yes	Customer note

Table 4.6 Order Database Table.

8. Payment Table: This table consists of 19 fields with data types including bigint, varchar, decimal, datetime, and tinyint. The 'id' field serves as the primary key and is set to auto-increment. Other specializations include form_id, status, subtotal_amount, discount_amount, total_amount, currency, entry_id, gateway, type, mode, transaction_id, customer_id, subscription_id, subscription_status, title, date_created_gmt, date_updated_gmt, and is_published, which accommodate various data ranging from monetary values to identifiers and status information.

No.	Field Name	Data Type	Null	Description
1	id	bigint	No	Unique ID
2	form_id	bigint	No	Form ID
3	status	varchar(10)	No	Form status
4	subtotal_amount	decimal(26,8)	No	Subtotal
5	discount_amount	decimal(26,8)	No	Discount
6	total_amount	decimal(26,8)	No	Total
7	currency	varchar(3)	No	Currency code
8	entry_id	bigint	No	Entry ID
9	gateway	varchar(20)	No	Payment gateway
10	type	varchar(12)	No	Payment type
11	mode	varchar(4)	No	Processing mode

12	transaction_id	varchar(40)	No	Transaction ID
13	customer_id	varchar(40)	No	Customer ID
14	subscription_id	varchar(40)	No	Subscription ID
15	subscription_status	varchar(10)	No	Subscription status
16	title	varchar(255)	No	Item title
17	date_created_gmt	datetime	No	GMT creation date
18	date_updated_gmt	datetime	No	GMT update date
19	is_published	tinyint(1)	No	Publication status

Table 4.7 Payment Database Table.

Chapter 5 Technology Stack and Implementation

5.1 Selected Technology Stack

This technology stack is tailored to the needs of the Classified Multi-Vendor Ads Management Website. It encompasses front-end and back-end tools and frameworks for efficient development and functionality.[28]

5.1.1 Frontend Technologies

- **HTML/CSS/JavaScript:** These core technologies form the foundation for creating web pages. HTML structures content, CSS styles it for a polished appearance, and JavaScript adds interactivity and dynamic features.
- **Bootstrap:** The most popular CSS framework that makes constructing flexible and mobile-first web apps easier while guaranteeing a unified and user-friendly aesthetic.

5.1.2 Backend Technologies

- **PHP:** As the primary server-side scripting language, PHP handles the backend logic of the website. PHP is chosen for its versatility, making it accessible for newcomers while offering advanced features for experienced developers.
- **Laravel:** A robust PHP framework following the Model-View-Controller (MVC) architectural paradigm. For routing, authentication, sessions, caching, and other crucial web development activities, Laravel offers a variety of features with a clear and beautiful syntax.

5.1.3 Database Technologies

- **MySQL:** This widely adopted relational database management system is a suitable companion for PHP and is well-suited for web-based applications. It offers efficient data management capabilities and strong support for complex queries.
- **phpMyAdmin:** A free and open-source administration tool designed for MySQL and

MariaDB. phpMyAdmin simplifies database management through a web-based interface, facilitating various operations on the database, such as table creation, data insertion, and query execution.

5.2 Frontend Development and Implementation

The Classified Multi-Vendor Ads Management Website's front end was painstakingly created utilizing HTML, CSS, and Bootstrap, integrating these technologies to produce a simple yet visually appealing user experience.

HTML: The fundamental building element for the front end, HTML (Hypertext Markup Language), establishes the format and content of online pages. It makes it easier to create different components, including headers, forms, buttons, and tables. The website's material is organized and presented clearly and coherently, thanks primarily to HTML.

CSS: The styling and visual design of the front end are handled through CSS (Cascading Style Sheets). It controls the website's design, including the typeface, color scheme, and layout. Because CSS allows for customization, a consistent and aesthetically pleasant user experience is ensured across various devices and screen sizes.

Bootstrap: A prominent front-end framework called Bootstrap enhances front-end development by providing a large selection of pre-built elements and responsive grid systems. Making a responsive and mobile-friendly design is made more accessible as a result. Bootstrap streamlines front-end development by fostering uniformity and flexibility and reducing the need for heavy custom CSS coding.

The front end of the Classified Multi-Vendor Ads Management Website's primary goal is to provide users with an easy-to-use interface. It focuses on usability so visitors can easily browse the website, look at listings, communicate with sellers, and find relevant information.

5.2.1 Frontend Code

Here are a few codes that I've shared to help you to understand.

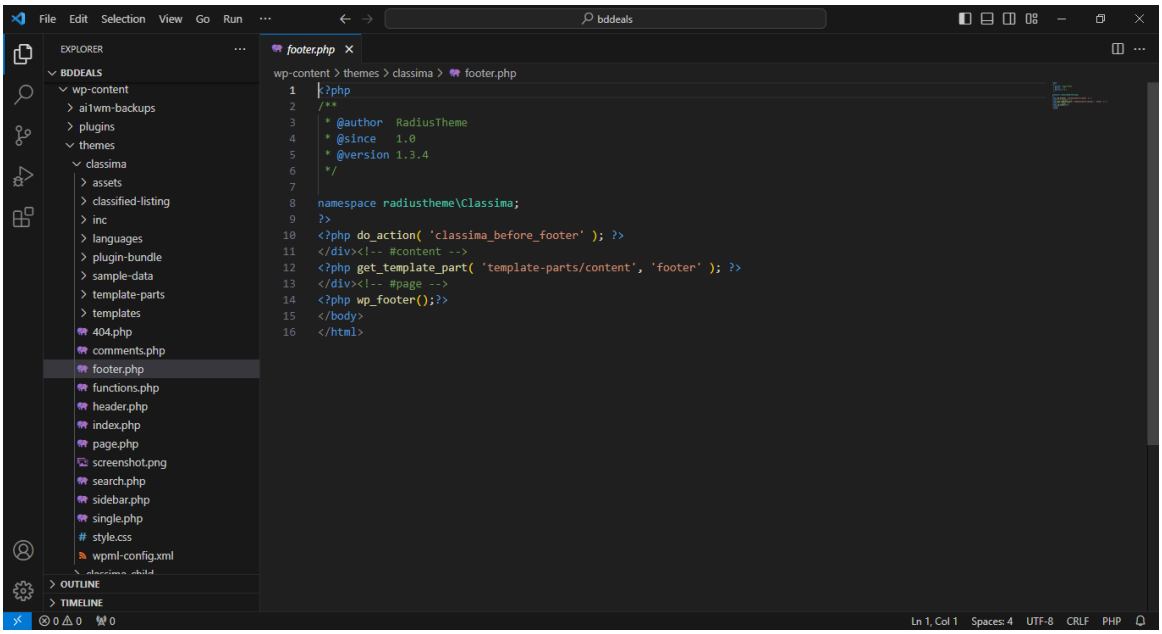


Figure 5.1 footer.php

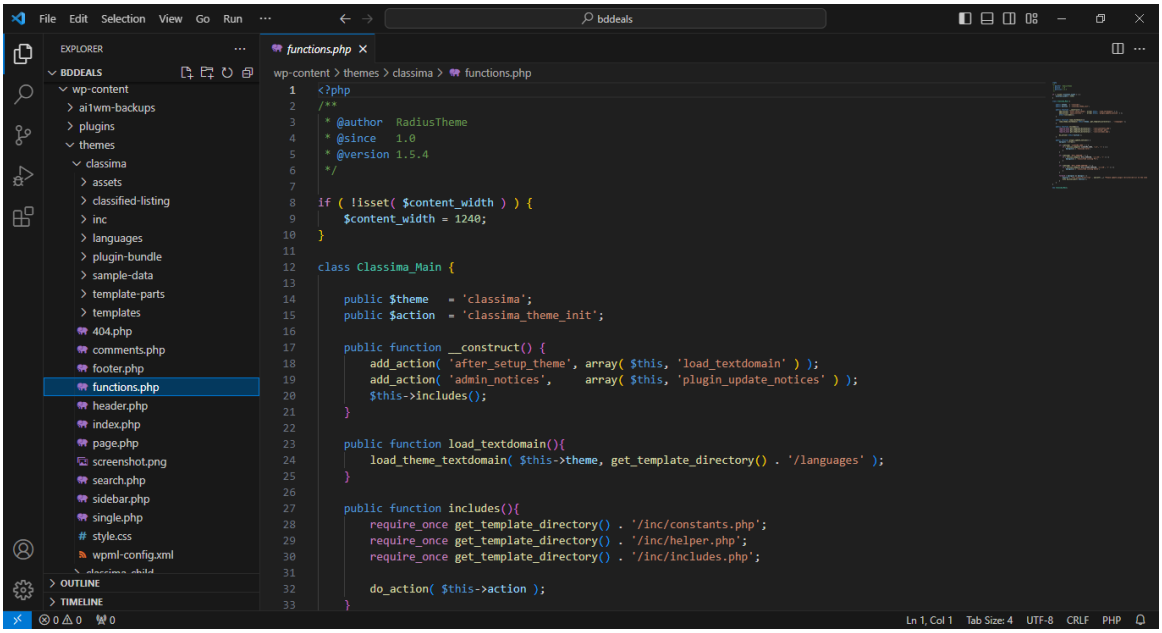


Figure 5.2 function.php

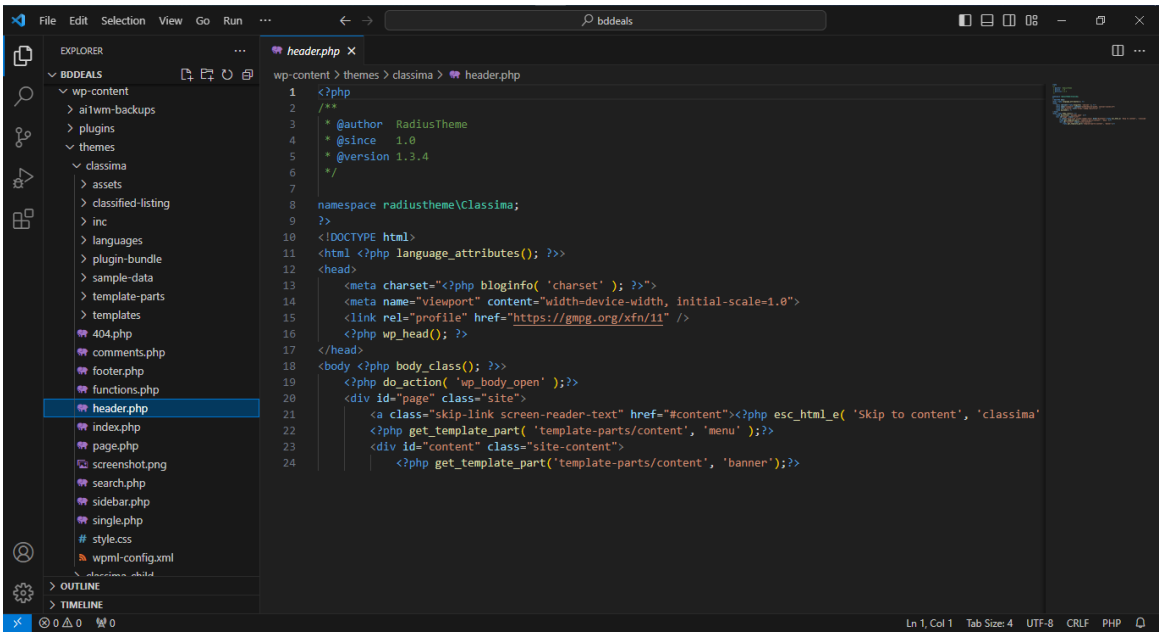


Figure 5.3 header.php

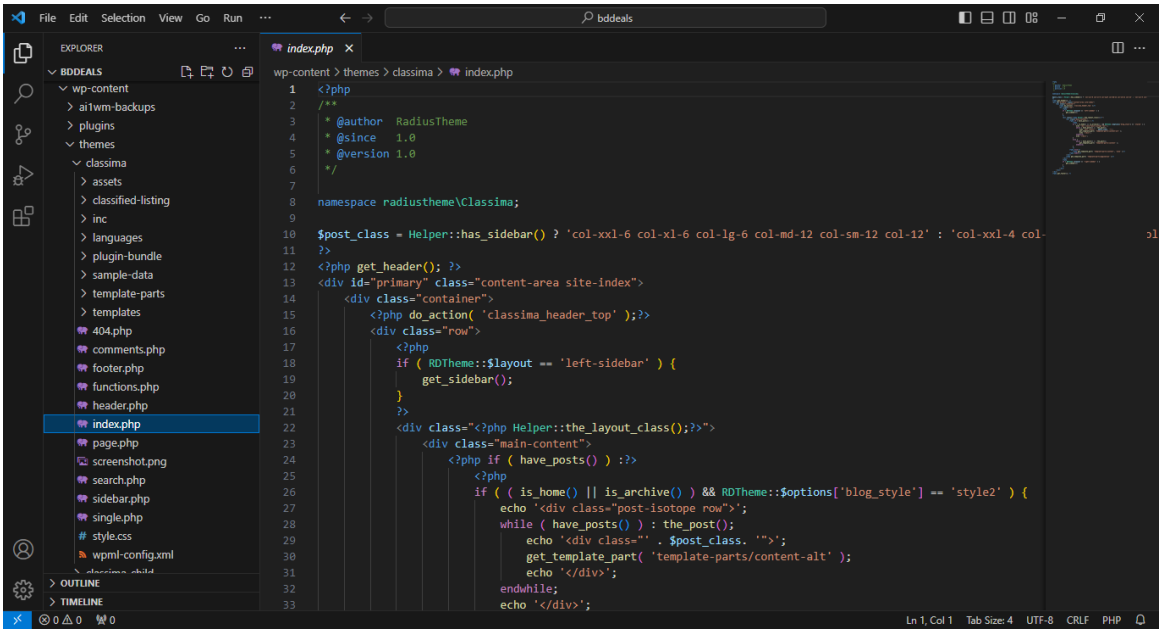


Figure 5.4 index.php

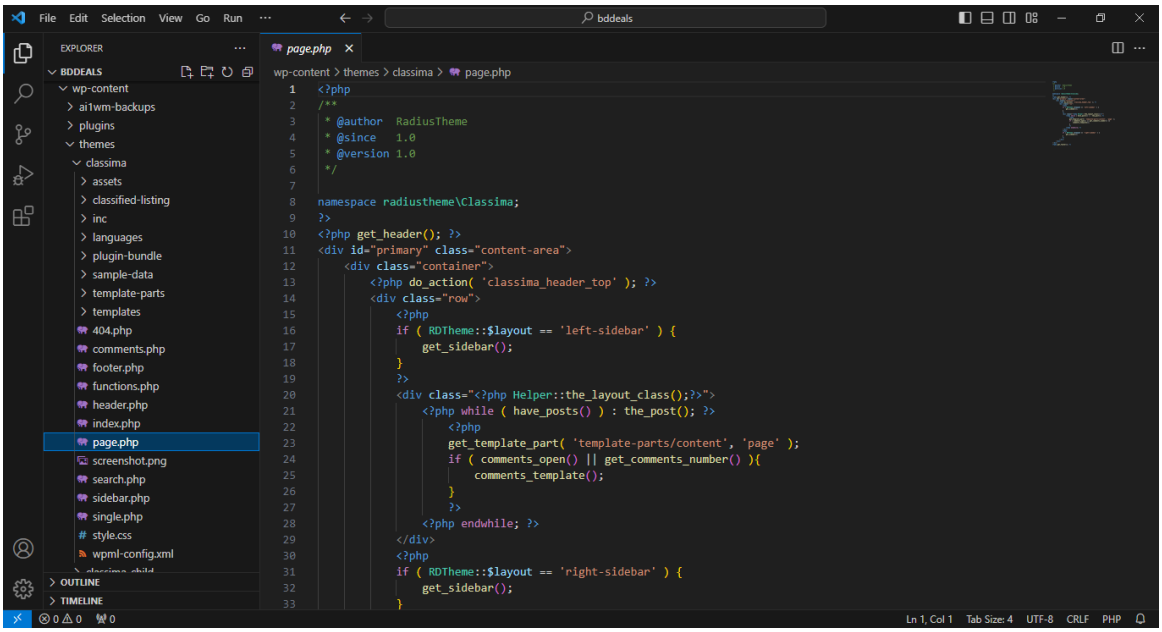


Figure 5.5 page.php

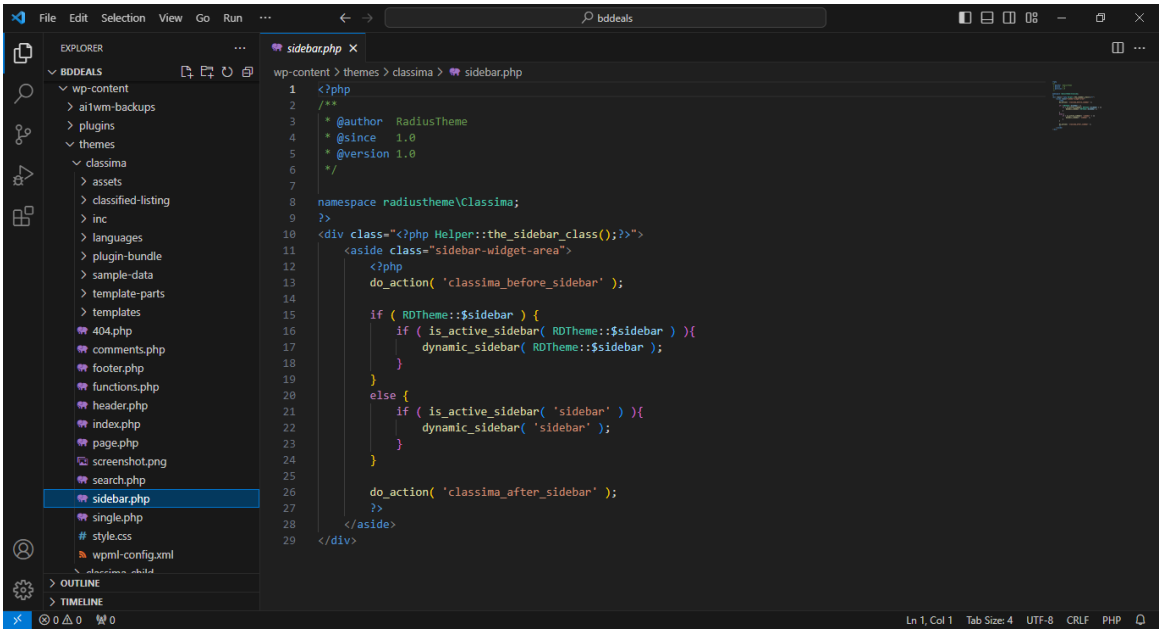


Figure 5.6 sidebar.php

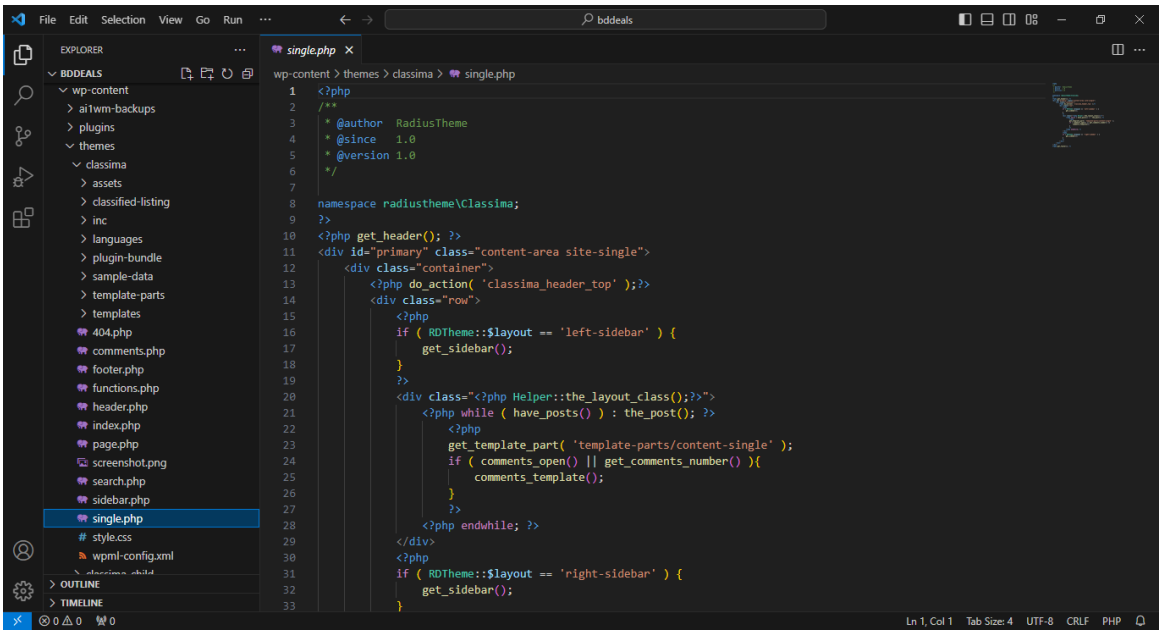


Figure 5.7 single.php

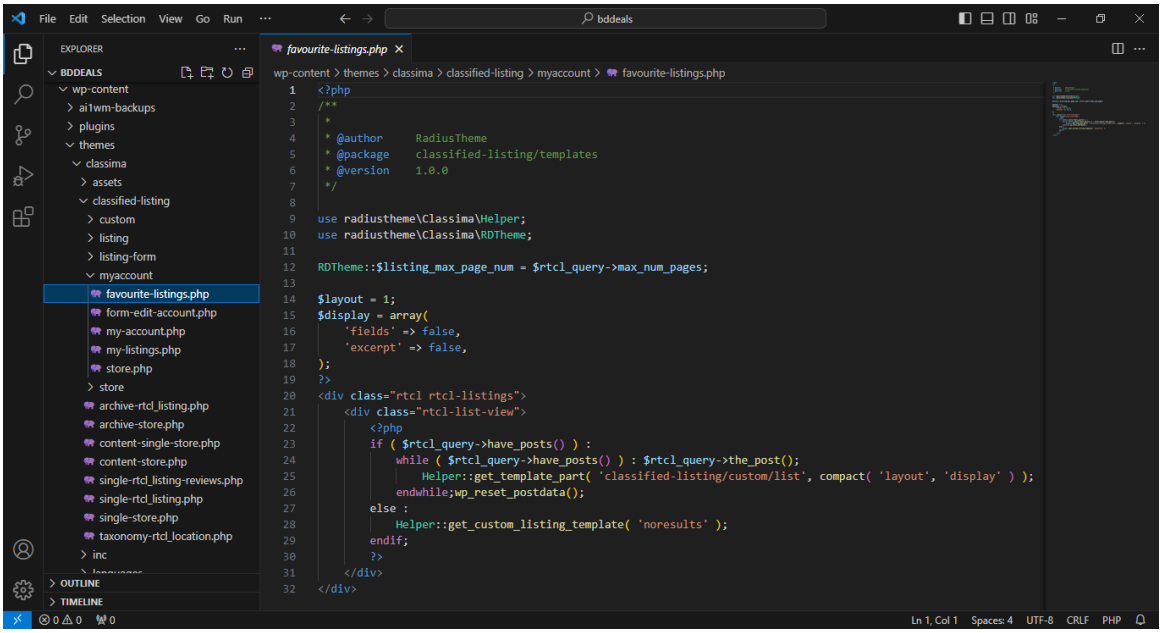


Figure 5.8 favourite-listing.php

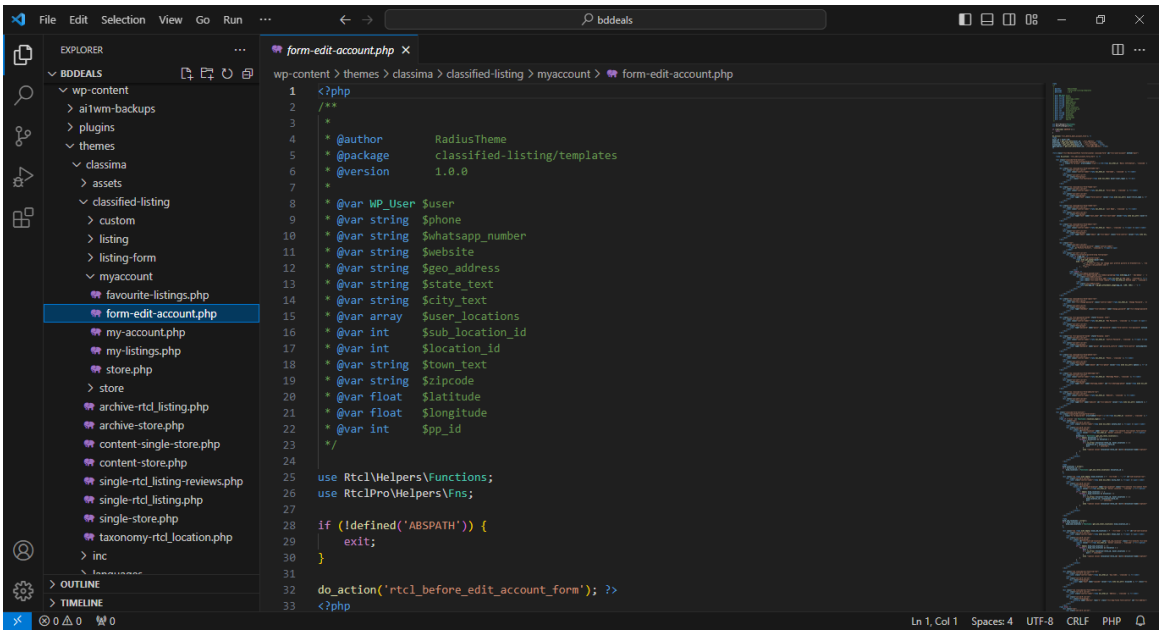


Figure 5.9 form-edit-account.php

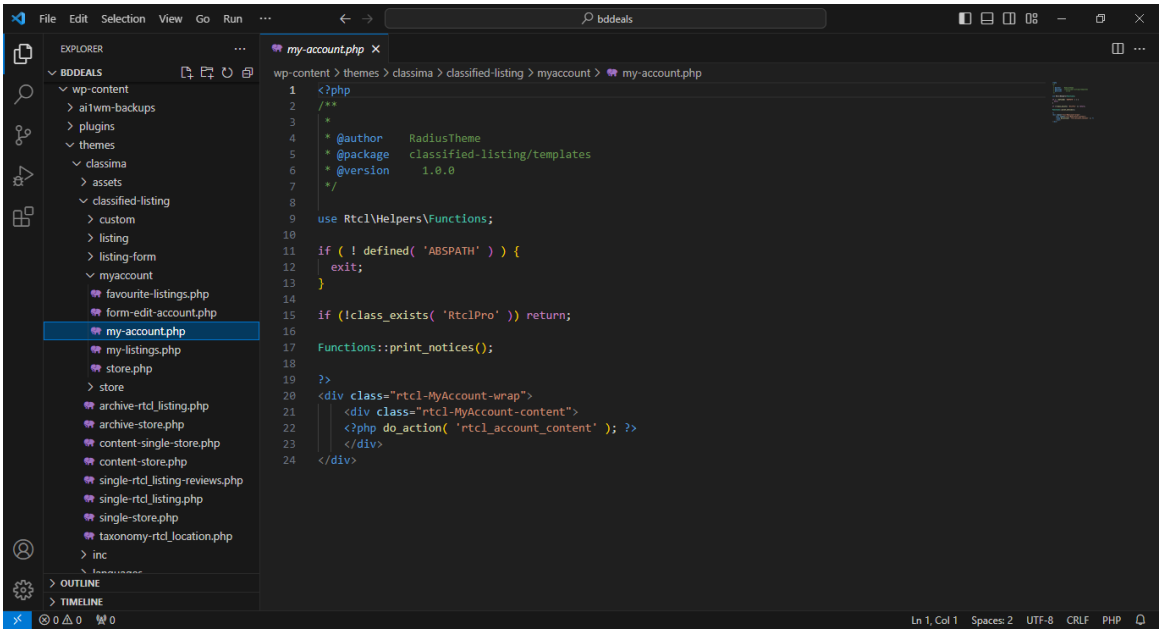


Figure 5.10 my-account.php

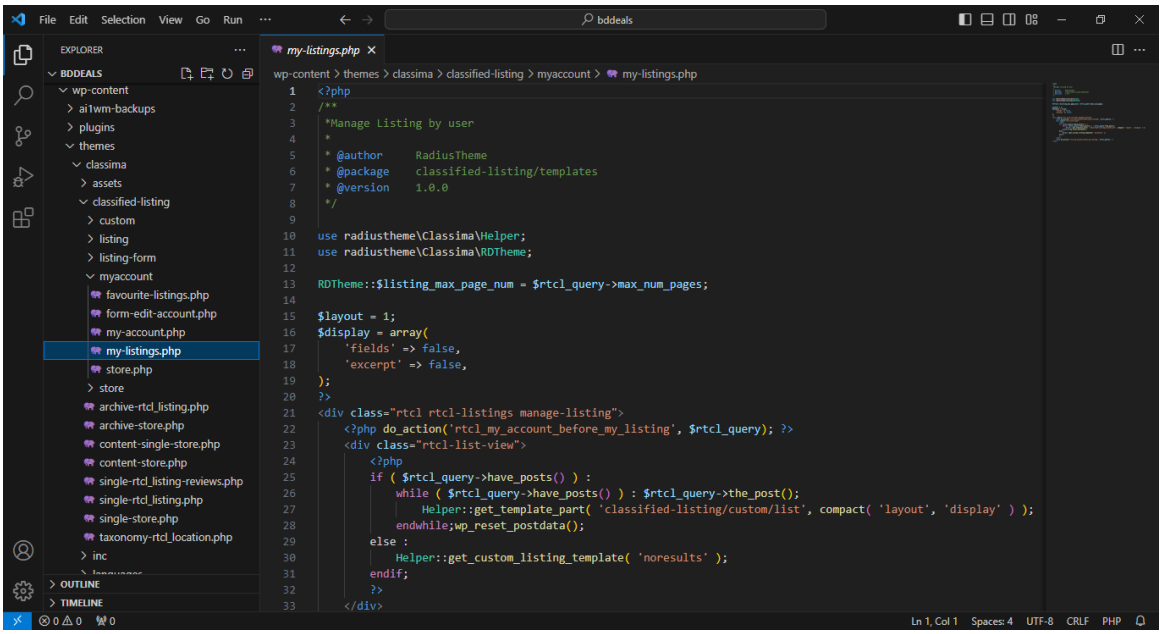


Figure 5.11 my-listings.php

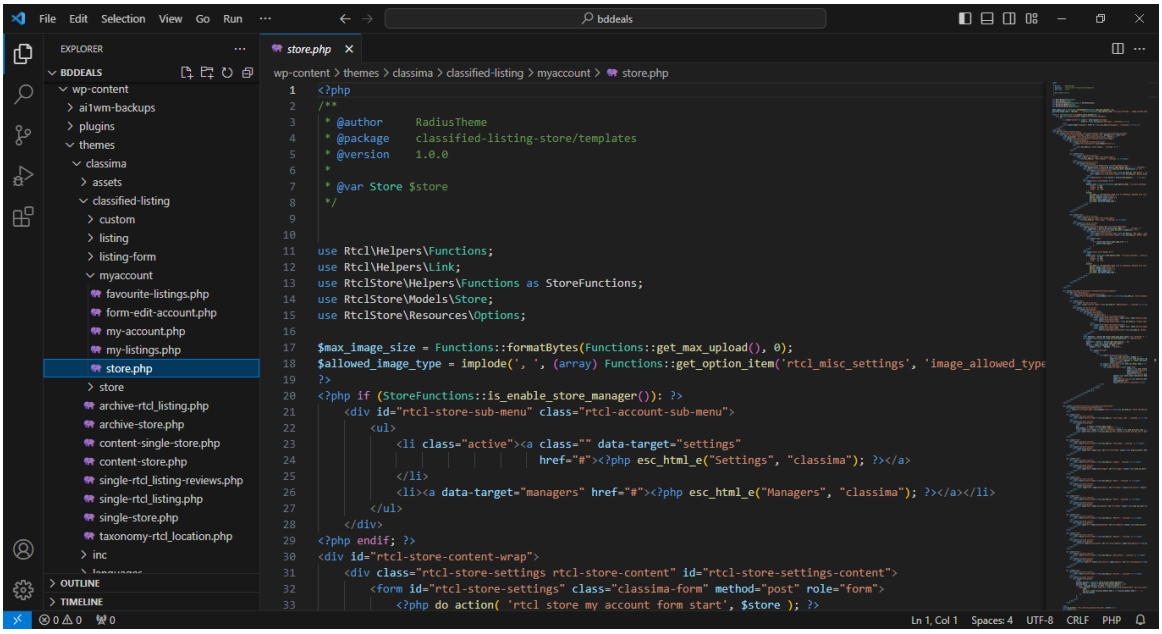


Figure 5.12 store.php

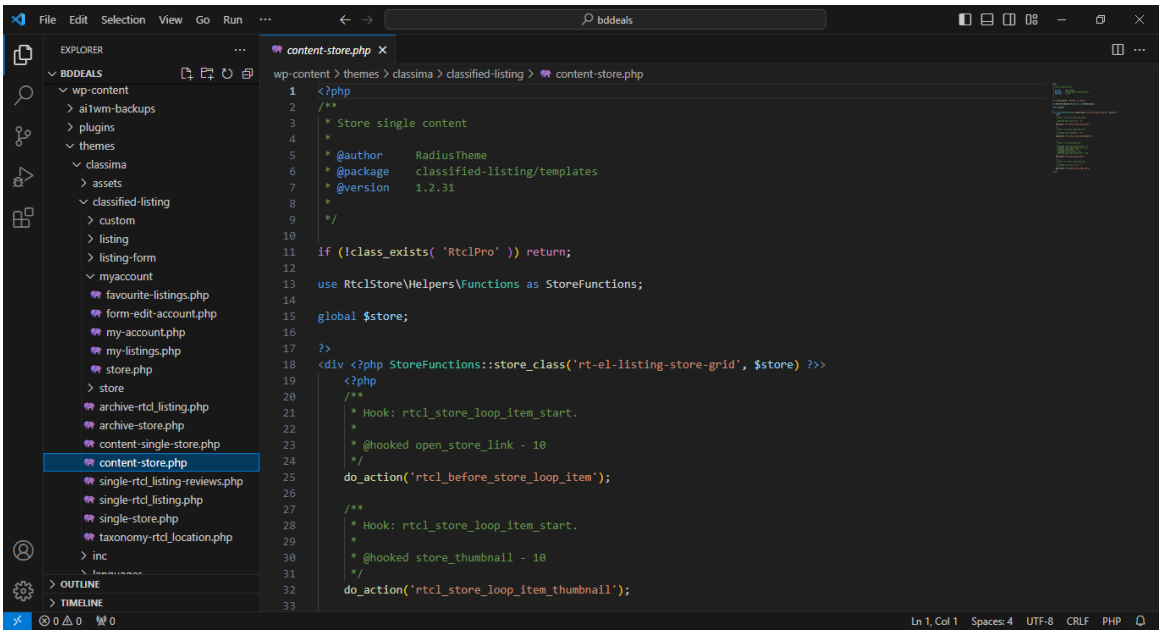


Figure 5.13 content-store.php

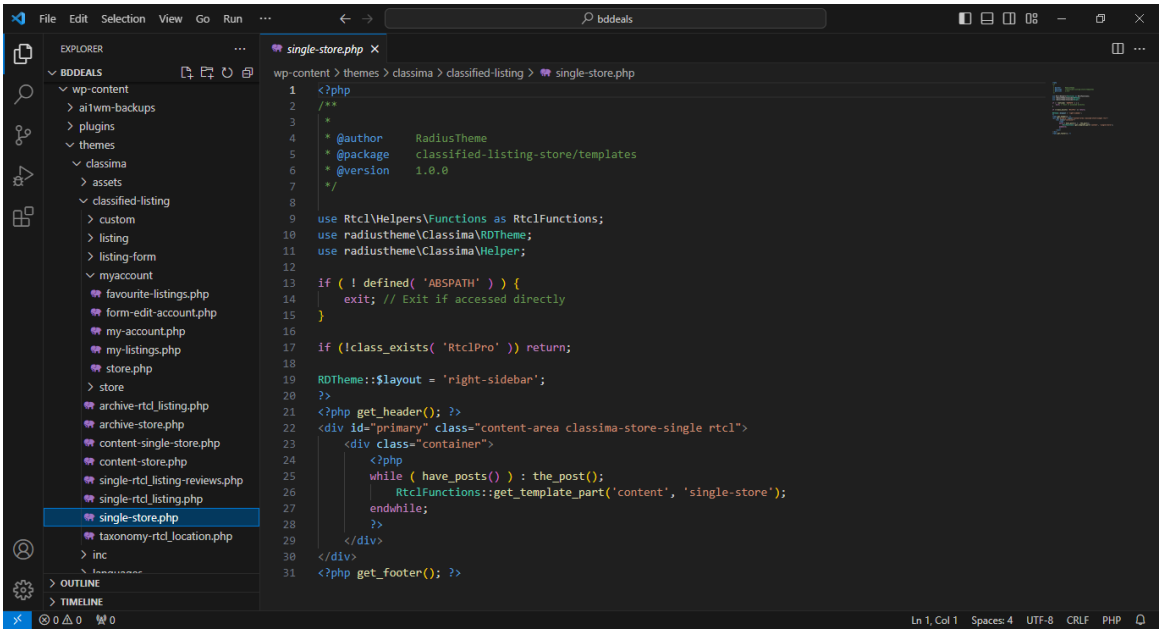


Figure 5.14 single-store.php

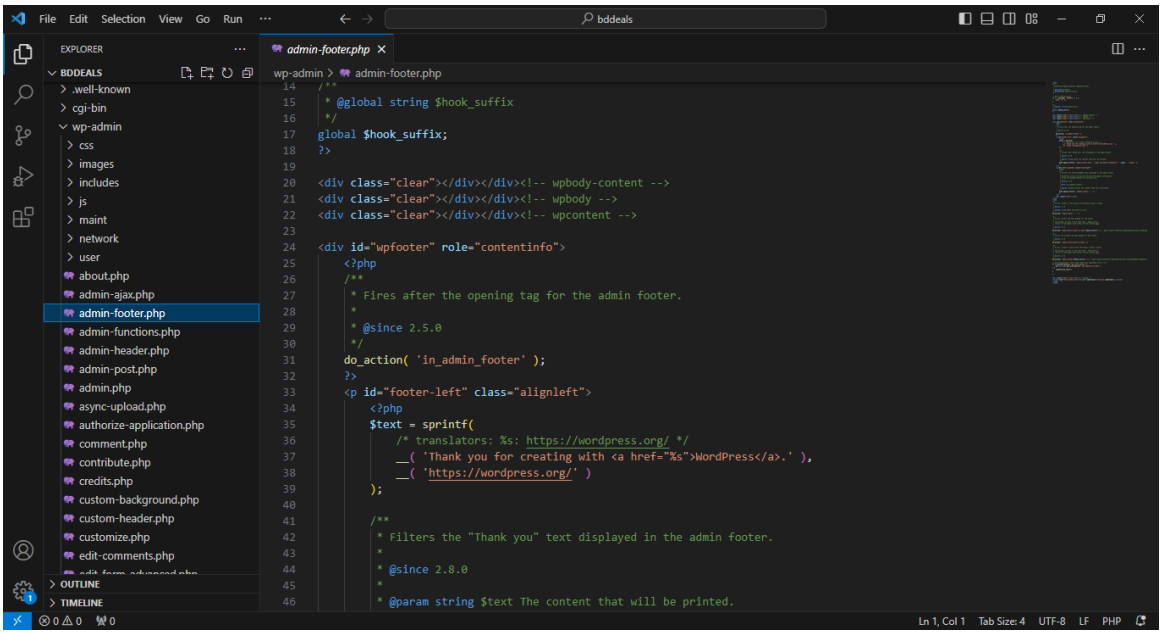


Figure 5.15 admin-footer.php

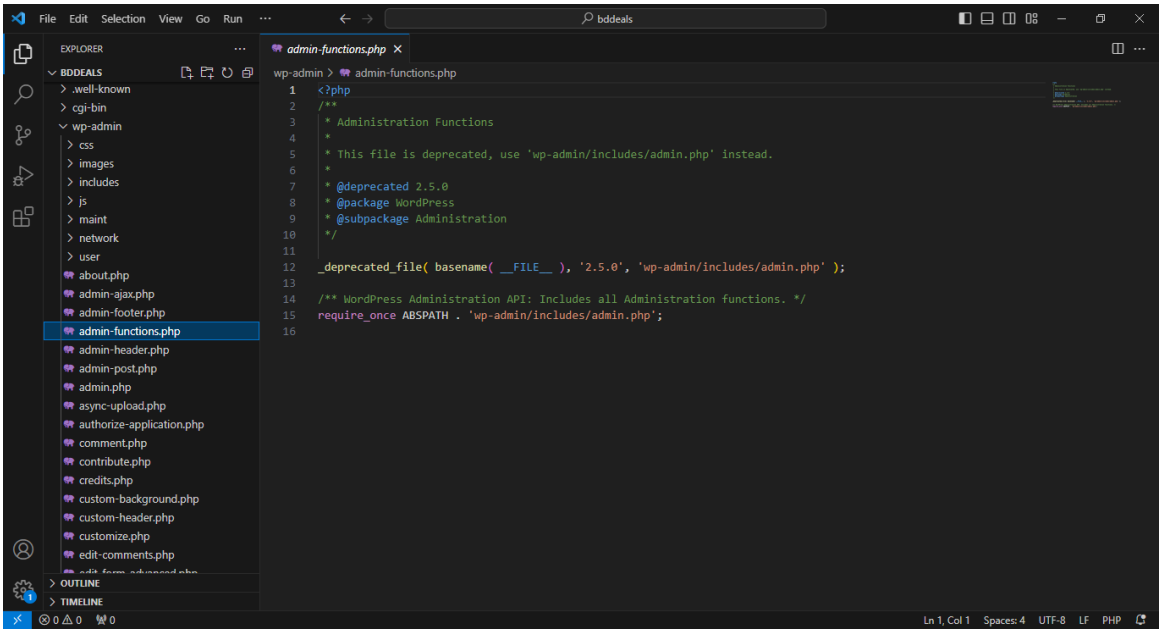


Figure 5.16 admin-function.php

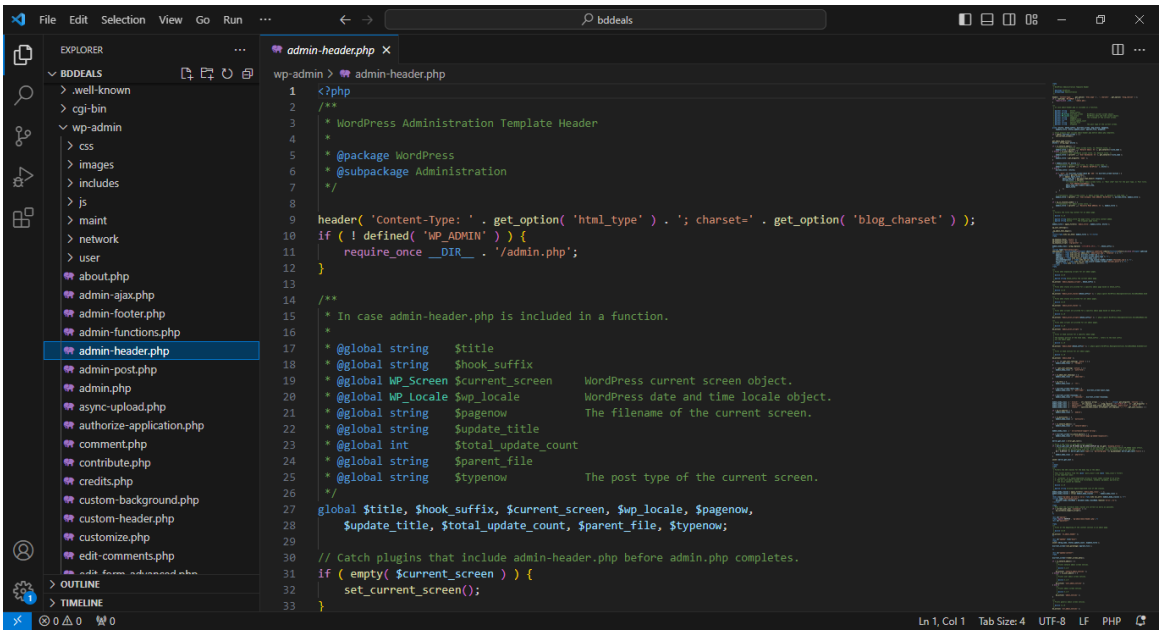


Figure 5.17 admin-header.php

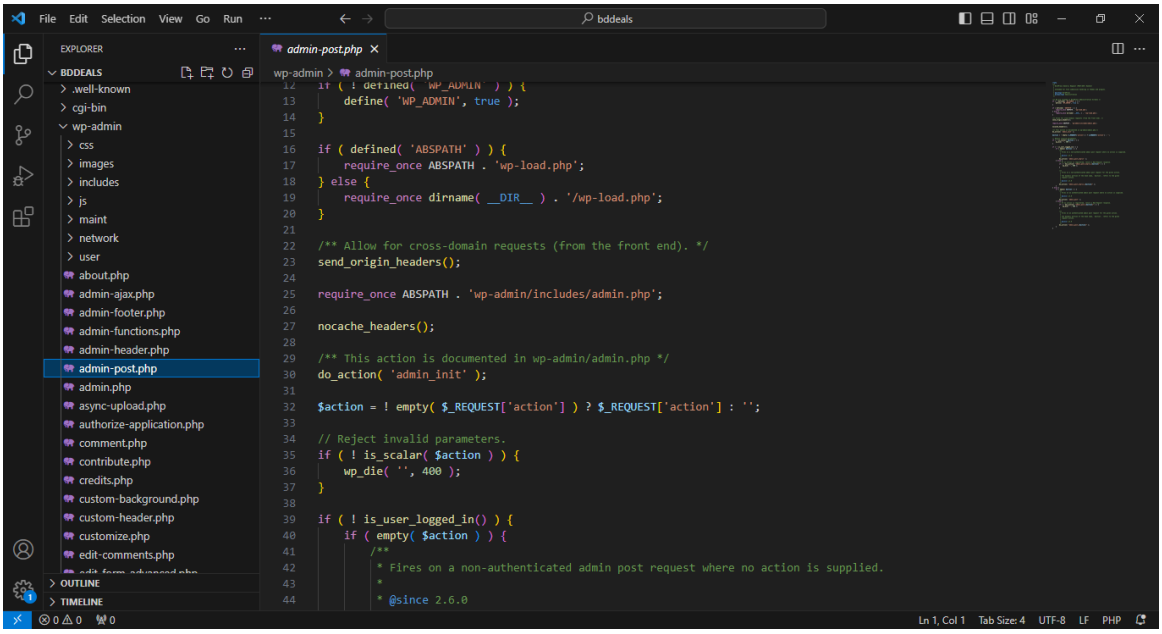


Figure 5.18 admin-post.php

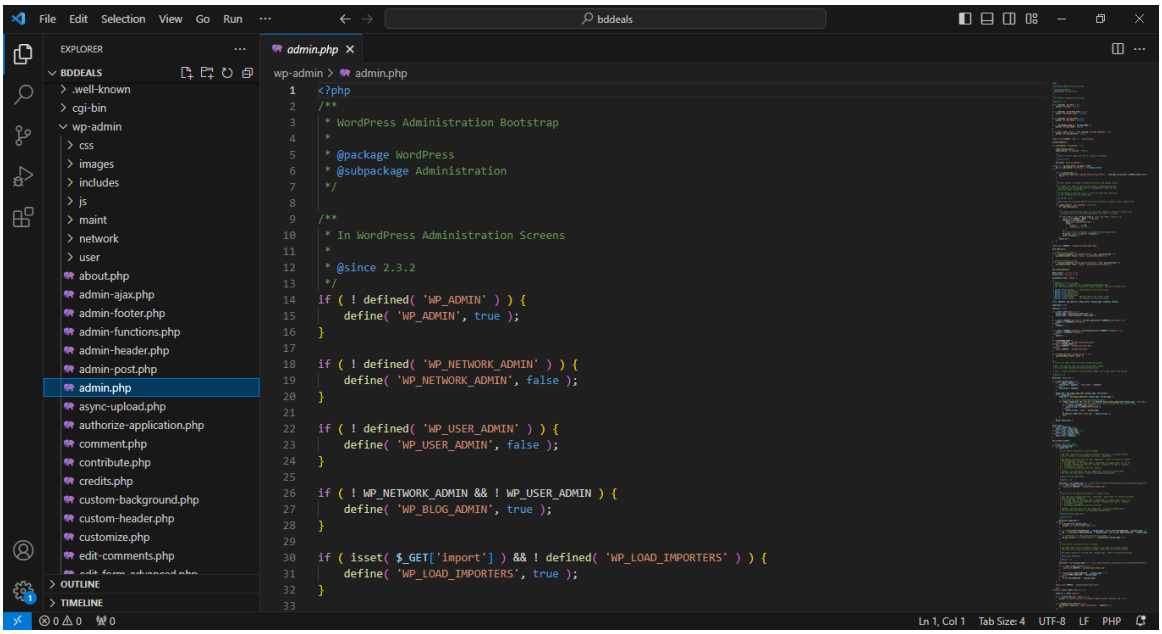


Figure 5.19 admin.php

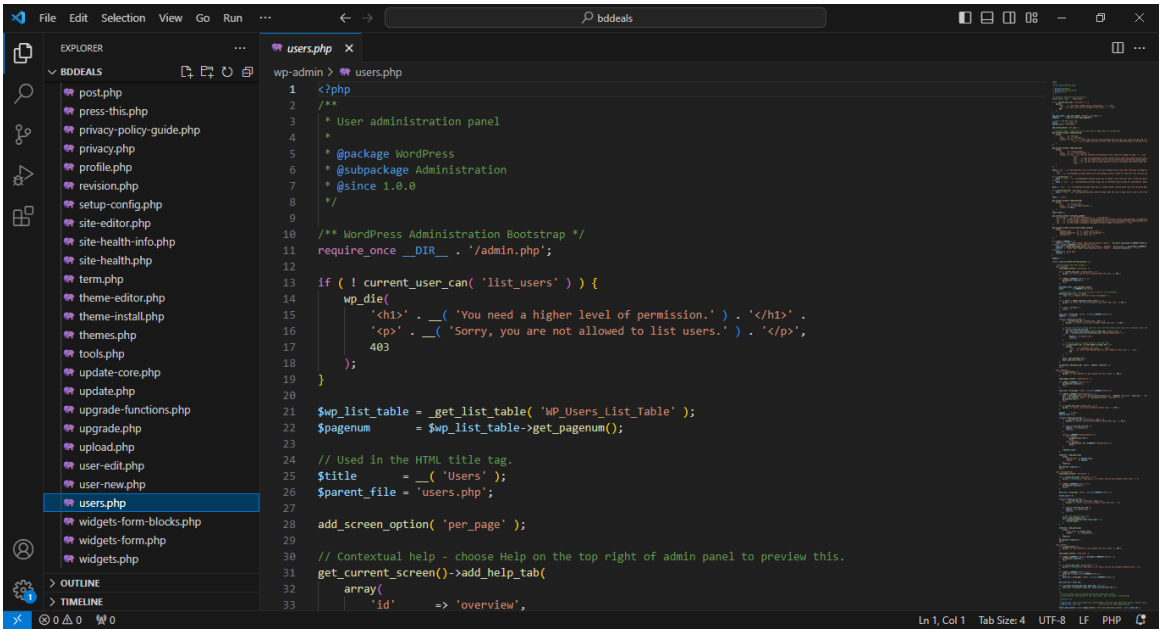


Figure 5.20 user.php

5.3 Backend Development and Implementation

PHP is a powerful and popular programming language, and the Laravel framework is used to build the backend of the Classified Multi-Vendor Ads Management Website.[29] The backend architecture of the website is built on a solid and practical foundation created by this strategic combination. PHP offers a wide range of capabilities that are perfect for web development. It is exceptional at handling database interactions, data processing, and applying complex business logic. It is a preferred option among web developers due to its broad library support and active development community.

Laravel: A robust PHP framework, Laravel enhances the development process by providing a systematic and beautiful way to build web apps. Routing, caching, security features, and database abstraction are just a few of the crucial elements that Laravel offers. These features speed up development and improve the quality of the code. Essential functions like user identification and authorization, data administration, the execution of business logic, and the smooth interface with other services are handled by the backend of the Classified Multi-Vendor Ads Management Website. It ensures the system will run effectively, continuously, and securely.

The backend development team can create an expandable and scalable system design by leveraging the capabilities of PHP and Laravel. While following coding best practices and industry standards, they may incorporate sophisticated features, enhance performance, and protect data integrity. The Classified Multi-Vendor Ads Management Website's PHP and Laravel backend allows the system to carry out a wide range of tasks, including managing vendor listings, handling user interactions, producing reports, and facilitating fluid communication between the front end and the database.

5.3.1 Backend Code

Here are a few codes that I've shared to help you to understand.

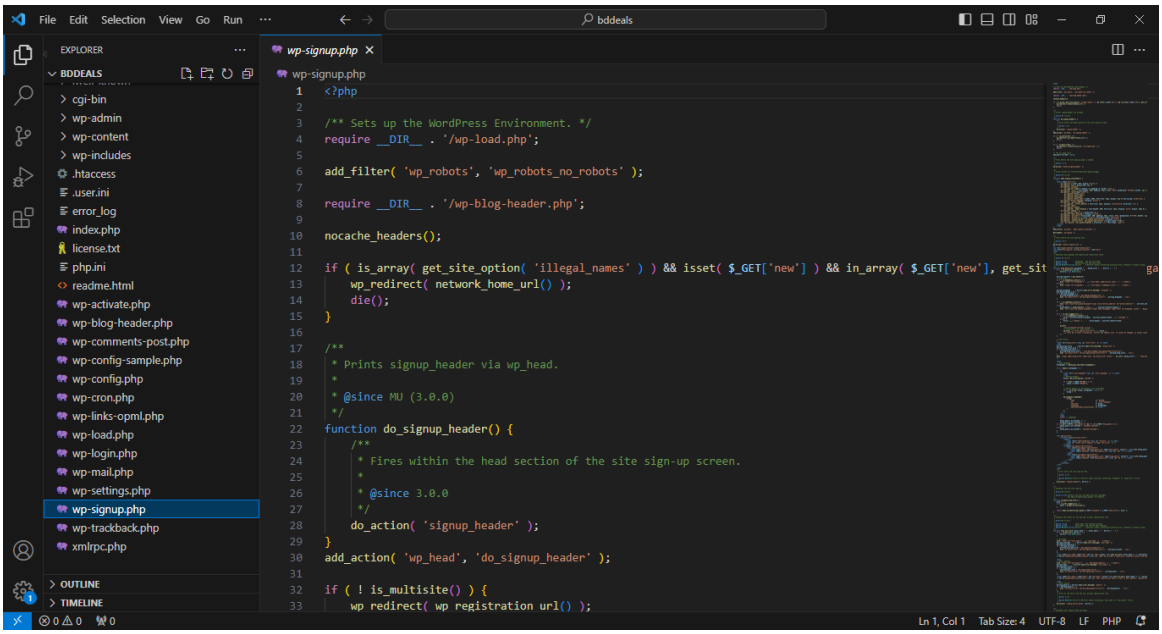


Figure 5.21 wp-signup.php

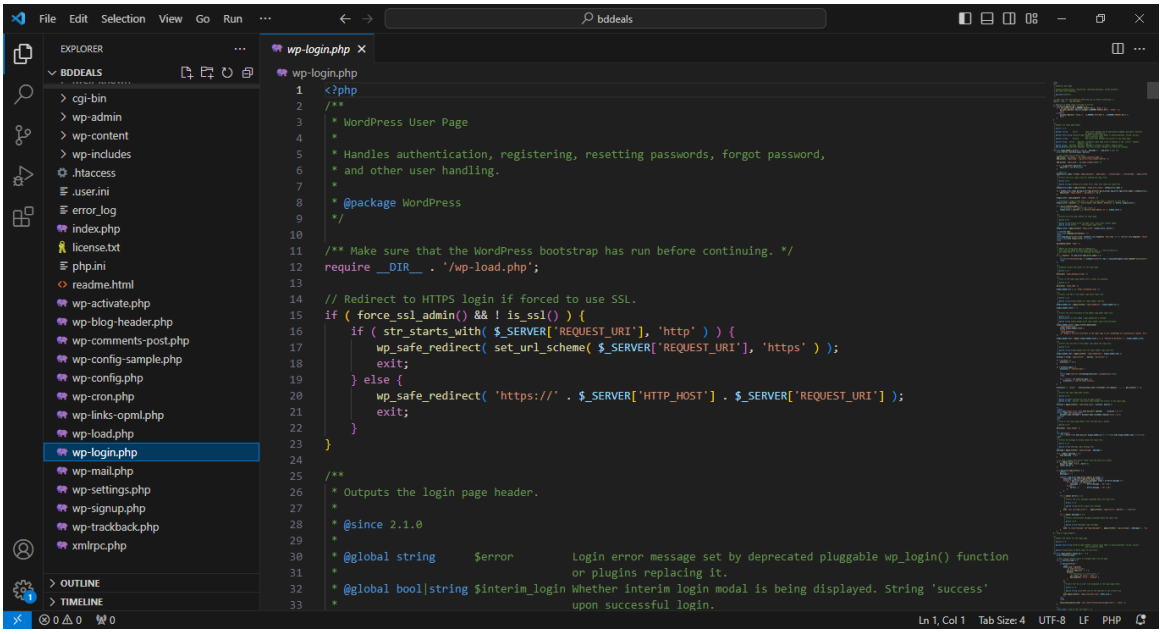


Figure 5.22 wp-login.php

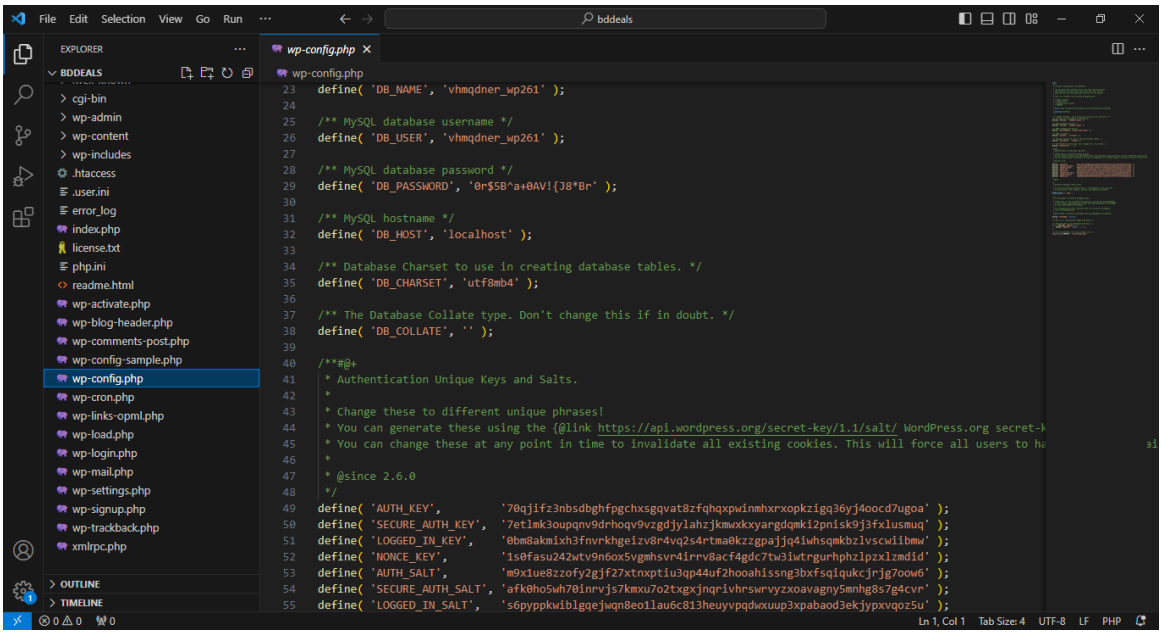


Figure 5.23 wp-config.php

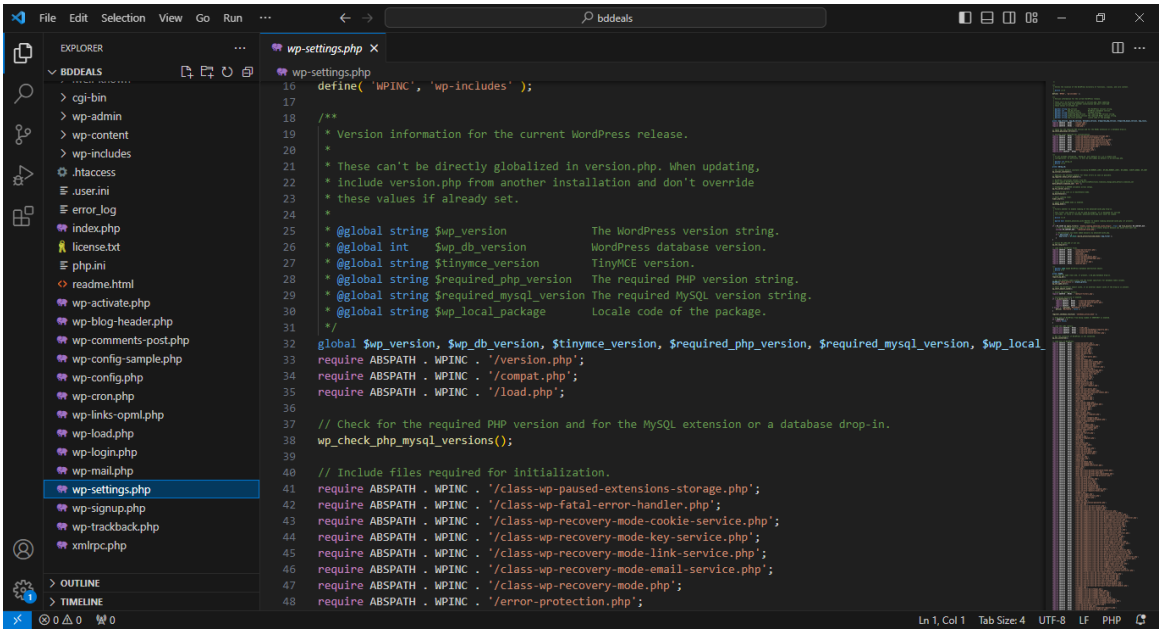


Figure 5.24 wp-settings.php

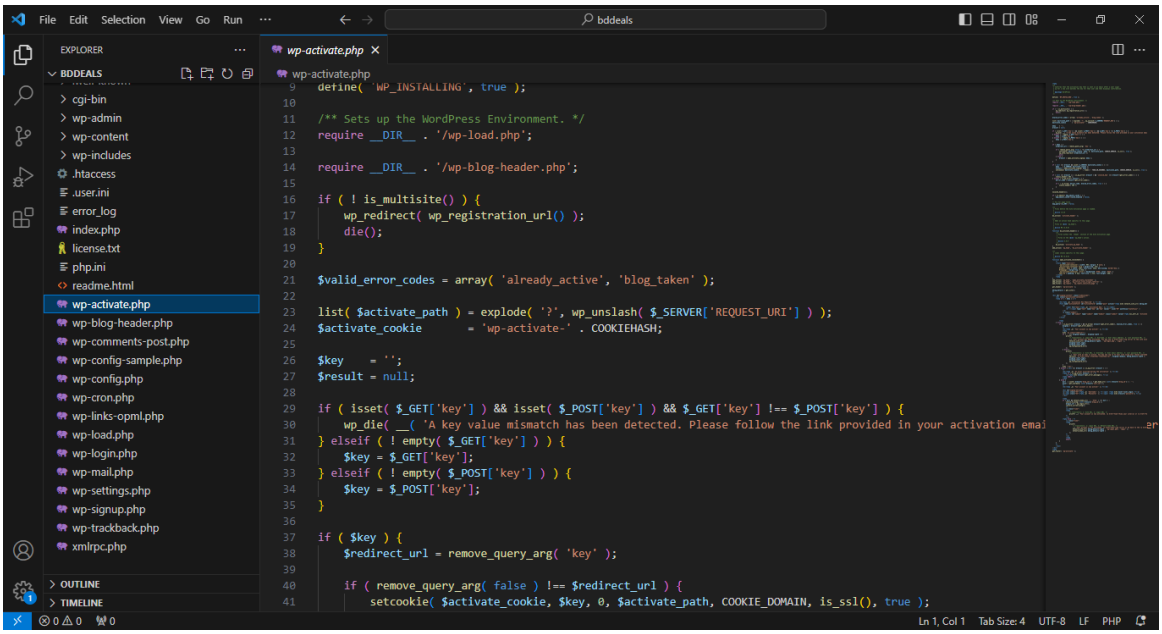


Figure 5.25 wp-activate.php

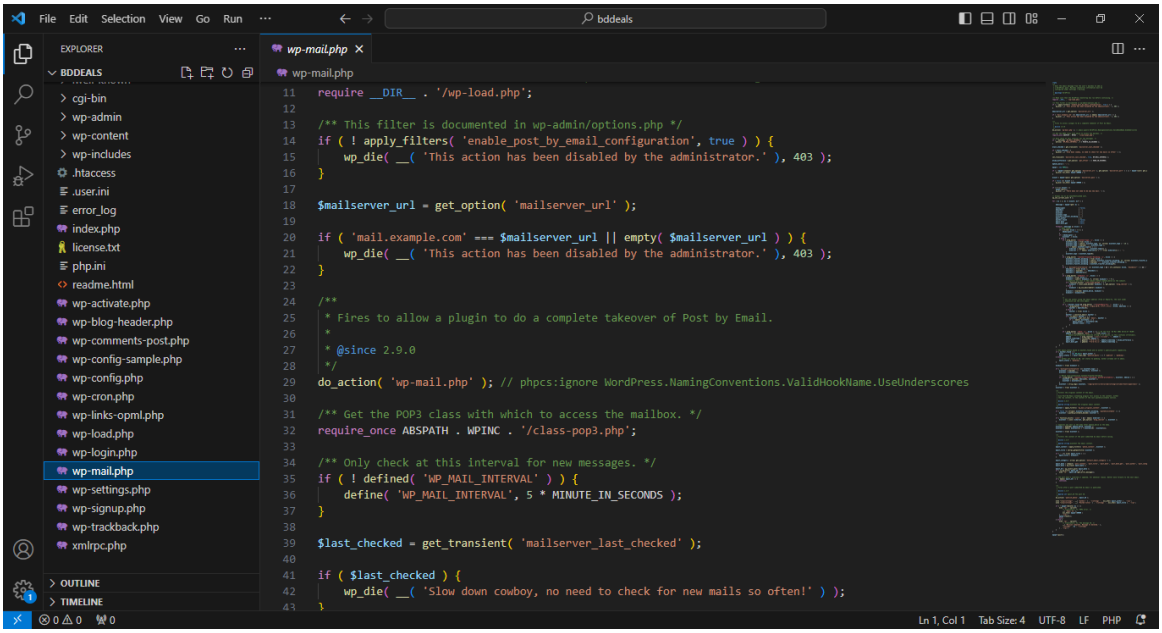


Figure 5.26 wp-mail.php

Chapter 6 System Testing and Report

6.1 Testing Strategy and Plan

This chapter details a thorough testing approach and plan for the Classified Multi-Vendor Ads Management Website, ensuring its robustness and dependability. Testing is vital in the software development lifecycle, ensuring the system achieves its goals, operates optimally, and maintains security.[30] The systematic procedure used is described in this section.

6.2 Functional Testing

The testing strategy primarily focuses on functional testing, which evaluates the website's compliance with its functional specifications. This stage involves rigorous validation of features, functions, and user interactions using boundary value analysis and equivalence partitioning methods. The results provide crucial insights into the system's correctness, precision, and responsiveness.

6.3 Performance Evaluation

The Classified Multi-Vendor Ads Management Website's performance evaluation is crucial for managing predicted and peak loads. Through load and stress testing, I identified bottlenecks and scalability issues. The methodology, outcomes, and suggestions for improvement are discussed in this section.

6.4 Usability Assessment

The usability assessment evaluates the website's user-friendliness and overall experience by conducting usability testing with end-users.[31] The process involves gathering feedback on navigation, layout, and accessibility. The results are used to identify areas for improvement in the user interface.

6.5 Security Testing

The classified advertisements management website is crucial for managing customer data and transactions. Extensive security testing was conducted to identify flaws, evaluate data protection protocols, and ensure compliance with industry requirements. This section details the methods used, flaws found, and actions taken to reduce risks.

6.6 User Acceptance Testing (UAT)

User acceptance testing (UAT) is the final stage of testing, where end users evaluate a system's readiness for production deployment. A representative group of users evaluates the website in real-world scenarios, providing insights into the UAT process, user input, and changes in response to user ideas.

6.7 Testing Results and Comprehensive Report

No.	Test	Result (Expected)	Result (Actual)	Status
1	Login	No Login	No Login	Passed
2	Login	Login	Login	Passed
3	Account Create	Created	Created	Passed
4	Membership	Purchased	Purchased	Passed
5	Payment	Completed	Completed	Passed
6	Security	Access Denied	Access Denied	Passed
7	Website Hosted	Running	Running	Passed

Table 6.1 Testing Results

Conclusion

Conclusion:

The Classified Multi-Vendor Ads Management Website is a dynamic platform that combines contemporary technology and user-centric design. It is a cornerstone in online classified ads, supported by a structured database. The complex but user-friendly interface offers a challenging experience for users, sellers, and marketers, ensuring the project successfully meets its objectives. The Classified Multi-Vendor Ads Management Website is a user-friendly platform that simplifies ad management, fostering trust and openness in the online market. It enables easy exploration, posting, and management of ads while maintaining data integrity. This platform offers a strong foundation for future innovations and promises further growth in the dynamic world of online ads, making it a valuable resource for sellers and potential customers.

Future Scope:

In its trajectory of evolution, the Classified Multi-Vendor Ads Management Website is poised for a compelling and promising future characterized by a genuine commitment to continuous innovation and the refinement of the user experience. This expanded vision delineates the forthcoming scope of the website, addressing critical aspects for further development.

1. Payment Gateway Integration: In an ever-evolving online transaction landscape, strategically integrating a diversified array of payment gateways is paramount. This initiative is envisaged to cater to a broader user demographic while enhancing trust, security, and transactional convenience.

2. AI Integration: The website is strategically positioning itself to harness the potential of cutting-edge AI technologies. This integration will manifest in sophisticated recommendation engines, AI-driven customer support chatbots, and intelligent data analytics. These AI-driven enhancements aim to provide users with a more personalized, efficient, and engaging experience.

3. Ads Optimization: An unwavering commitment to refining ad placement and targeting strategies is integral to the website's future. The platform will optimize ad content by employing

AI and data analytics to align with individual user preferences. This approach seeks to elevate the effectiveness of advertisements for vendors and make them less intrusive for users.

4. User Optimization: User-centricity remains paramount in the website's strategic outlook. The platform's evolution will be driven by user feedback and data analytics insights. These insights will inform refinements to the user interface, rendering it more intuitive and personalized. Furthermore, the incorporation of features such as mobile number login will serve to streamline the user experience.

5. Mobile Apps: Specialized mobile applications will be developed to cater to the burgeoning population of mobile device users. These applications will offer enhanced features and real-time notifications, enhancing the ease of interaction with the platform for users on smartphones and tablets.

6. Global Expansion: With its future trajectory, the website will embark on a global expansion strategy, venturing into new markets and diversifying its user base. Localization initiatives will play a pivotal role in adapting the platform to diverse languages and cultural nuances, ensuring a welcoming and culturally relevant user experience for a global audience.

7. Data Integrity and Security: Rigorous enhancements will be deployed to fortify user data protection, instilling user trust. Incorporating comprehensive data analytics will facilitate identifying and preventing potential security threats, thereby upholding the highest standards of data integrity.

The Classified Multi-Vendor Ads Management Website's future foresees boundless growth and innovation. With an unwavering commitment to technological progress, responsiveness to user feedback, and a steadfast pursuit of global expansion, the website aims to deliver an even more enriched and personalized experience to customers and vendors. Anchored by a strategic focus on payment gateway integration, AI-driven innovation, ads optimization, user optimization, mobile applications, and stringent security measures, the platform is well-positioned to thrive in the dynamic and perpetually evolving landscape of the digital classifieds market.

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[9] https://www.locanto.cn/ is referred to as a popular multi-vendor web application in China.

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[17] https://www.ebay.com/ is a popular multi-vendor web application in the USA and worldwide.

[18] https://www.alibaba.com/ is a popular multi-vendor web application in China and worldwide.

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